

partition function in Quantum case

$$Z = \sum_{n_1, n_2, \dots, n_N} e^{-\beta \sum_j \hbar \omega_j (n_j + \frac{1}{2})} = \prod_{j=1}^N \sum_{n_j=0}^{\infty} e^{-\beta \hbar \omega_j n_j} e^{-\beta \hbar \omega_j \frac{1}{2}}$$

$$= \prod_{j=1}^N \frac{1}{1 - e^{-\beta \hbar \omega_j}} e^{-\beta \hbar \omega_j \frac{1}{2}}$$

$\hbar \rightarrow 0$ approach
classical result of

$$\prod_j \frac{1}{\beta \hbar \omega_j}$$

$$F = + k_B T \sum_{j=1}^N \left[\ln(1 - e^{-\beta \hbar \omega_j}) + \frac{1}{2} \beta \hbar \omega_j \right]$$

$$\ln(1+x) = x$$

$$S = - \frac{\partial F}{\partial T} = - k_B \sum_{j=1}^N \left[\ln(1 - e^{-\beta \hbar \omega_j}) + \frac{1}{2} \beta \hbar \omega_j \right]$$

$$- k_B \sum_{j=1}^N \left[\frac{-e^{-\beta \hbar \omega_j} \times \frac{\hbar \omega_j}{k_B T^2}}{1 - e^{-\beta \hbar \omega_j}} + \frac{1}{2} \frac{\hbar \omega_j}{k_B T} \right]$$

$$T \rightarrow 0$$

$$\beta \rightarrow \infty$$

$$S = - k_B \left[-e^{-\beta \hbar \omega_j} + \frac{1}{2} \beta \hbar \omega_j \right]$$

$$- \frac{\hbar \omega_j}{k_B T} e^{-\beta \hbar \omega_j} + \frac{1}{2} \beta \hbar \omega_j$$

$$= k_B \left(\left(1 + \frac{\hbar \omega_j}{k_B T}\right) e^{-\frac{\hbar \omega_j}{k_B T}} \right) \rightarrow 0$$

$$U = - \frac{\partial \ln Z}{\partial \beta} = + \frac{\partial}{\partial \beta} \sum_{j=1}^N \left\{ \ln(1 - e^{-\beta \hbar \omega_j}) + \beta \hbar \omega_j \frac{1}{2} \right\}$$

$$= \sum_{j=1}^N \left\{ \frac{e^{-\beta \hbar \omega_j} (\hbar \omega_j)}{1 - e^{-\beta \hbar \omega_j}} + \hbar \omega_j \frac{1}{2} \right\}$$

$$f_j = \frac{1}{e^{\beta \hbar \omega_j} - 1}$$

$$= \sum_j \hbar \omega_j \left(f_j + \frac{1}{2} \right)$$

$$= \sum_{j=1}^N \left\{ \frac{\hbar \omega_j}{e^{\beta \hbar \omega_j} - 1} + \hbar \omega_j \frac{1}{2} \right\}$$

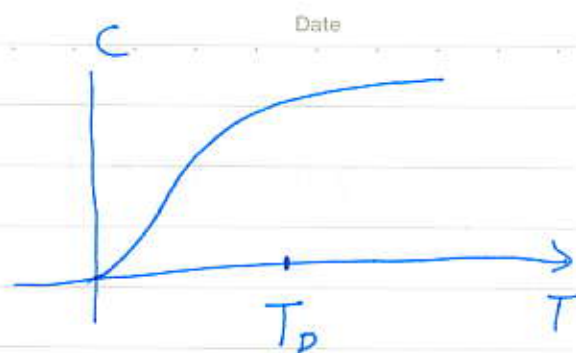
$$\hbar \rightarrow 0$$

$$= \sum_{j=1}^N \left\{ \frac{1}{\beta} + \hbar \omega_j \frac{1}{2} \right\} = N k_B T$$

$$C = \frac{dU}{dT} = \sum_{j=1}^N \frac{\hbar \omega_j (-1)}{(e^{\beta \hbar \omega_j} - 1)^2} e^{\beta \hbar \omega_j} \left(-\frac{\hbar \omega_j}{k_B T^2} \right) \quad (\beta \hbar \omega_j = 8 \times)$$

$$= k_B \sum_{j=1}^N \frac{e^{x_j} x_j^2}{(e^{x_j} - 1)^2}$$

$$\frac{\hbar \omega_M}{k_B T_D} = O(1)$$



Debye Temperature

must apply in ball-m...

admitting... from

$$\frac{1}{\epsilon_0} = \frac{1}{\epsilon_0} + \frac{1}{\epsilon_0} + \frac{1}{\epsilon_0} + \frac{1}{\epsilon_0} + \dots = \frac{1}{\epsilon_0} + \frac{1}{\epsilon_0} + \frac{1}{\epsilon_0} + \frac{1}{\epsilon_0} + \dots$$

... ..

$$(2, 8) A_{96} = 3(8, 8) A_{96}$$

$$A_{96} = 8 \quad \frac{1}{8} = 8 \quad \frac{1}{8} = 8$$

$$\frac{1}{T_{96}} = \frac{1}{T_{96}} + \frac{1}{T_{96}} + \frac{1}{T_{96}} + \frac{1}{T_{96}} + \dots = \frac{1}{T_{96}} + \frac{1}{T_{96}} + \frac{1}{T_{96}} + \frac{1}{T_{96}} + \dots$$

$$(2, 8) A_{96} = 3(8, 8) A_{96}$$

$$\frac{1}{T_{96}} = \frac{1}{T_{96}} + \frac{1}{T_{96}} + \frac{1}{T_{96}} + \frac{1}{T_{96}} + \dots = \frac{1}{T_{96}} + \frac{1}{T_{96}} + \frac{1}{T_{96}} + \frac{1}{T_{96}} + \dots$$

$$\frac{1}{T_{96}} = \frac{1}{T_{96}} + \frac{1}{T_{96}} + \frac{1}{T_{96}} + \frac{1}{T_{96}} + \dots = \frac{1}{T_{96}} + \frac{1}{T_{96}} + \frac{1}{T_{96}} + \frac{1}{T_{96}} + \dots$$

Lebesgue

Euler-Maclaurin's integral formula

More precise formula

$$\ln n! = n \ln n - n + \frac{1}{2} \ln n + \ln \sqrt{2\pi} + \frac{1}{12n} - \frac{1}{360n^3} + \dots$$

Arfken, Mathematical methods for physicists

Ivan's Theorem

$$\int d\Gamma (A, B) C = \int d\Gamma A (B, C)$$

Let $A = e^{-\beta H}$ $B = B(t)$, $C = A$

note that $\frac{\partial}{\partial \beta} (B, A(t)) = \frac{\partial}{\partial H} (H, B(t)) = -\frac{1}{k_B T} (H, B(t))$

$$\frac{\partial}{\partial \beta} = \frac{\partial}{\partial H} \frac{\partial H}{\partial \beta} \quad \text{since } \frac{\partial}{\partial \beta}$$

prove $\int d\Gamma (A, B) C = \int d\Gamma A (B, C)$

consider $\int d\Gamma C \frac{\partial A}{\partial q_j} \frac{\partial B}{\partial p_j} = \int d\Gamma \left[\frac{\partial}{\partial q_j} \left(C A \frac{\partial B}{\partial p_j} \right) - A \frac{\partial C}{\partial q_j} \frac{\partial B}{\partial p_j} - A C \frac{\partial^2 B}{\partial q_j \partial p_j} \right]$

exchange $q \leftrightarrow p$ sum over j , we get the result

$\frac{\partial^2}{\partial q_j \partial p_j} = \frac{\partial^2}{\partial p_j \partial q_j}$

$$\oint C A \frac{\partial B}{\partial p}$$

surface term is assumed zero.

① Stirling formula

$$n! = 1 \cdot 2 \cdot 3 \cdots (n-1) \cdot n$$

$$\ln n! = \sum_{i=1}^n \ln i \approx \int_1^n \ln x \, dx$$

$$= x \ln x \Big|_1^n - \int_1^n x \, d \ln x$$

$$\int_1^n 1 \, dx = n-1$$

$$= n \ln n - n + 1$$

② $\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} \, dx = I$

$$I^2 = \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} \, dx \int_{-\infty}^{\infty} e^{-\frac{y^2}{2}} \, dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{x^2+y^2}{2}} \, dx \, dy$$

$$= \int_0^{\infty} e^{-\frac{r^2}{2}} r \, dr \, d\theta = 2\pi \int_0^{\infty} e^{-\frac{r^2}{2}} \frac{r^2}{2} \, d\frac{r^2}{2} = 2\pi$$

$$I = \sqrt{2\pi}$$

③ $\Omega_n = \int dX_1 dX_2 \cdots dX_n = C_n R^n$ $S_n = \frac{d\Omega_n}{dR} = n C_n R^{n-1}$

$$\sum x_i^2 < R^2$$

$$\pi^{\frac{n}{2}} = \int_{-\infty}^{\infty} dX_1 \cdots \int_{-\infty}^{\infty} dX_n e^{-X_1^2 + X_2^2 + \cdots + X_n^2} = \int_0^{\infty} dR S_n(R) e^{-R^2}$$

gaussian
 $R^2 = t$

$$= n C_n \int_0^{\infty} dR R^{n-1} e^{-R^2}$$

$$\Gamma(x+1) = x \Gamma(x)$$

$$= \frac{1}{2} n C_n \int_0^{\infty} dt t^{\frac{n}{2}-1} e^{-t} = \frac{1}{2} n C_n \Gamma\left(\frac{n}{2}\right)$$

$$\pi^{\frac{n}{2}} = \frac{1}{2} n \Gamma\left(\frac{n}{2}\right) C_n$$

$$= \Gamma\left(\frac{n}{2} + 1\right) C_n$$

$$R^2 = t$$



single particle states

Date

$$-\frac{\hbar^2}{2m} \nabla^2 \phi_r = E_r \phi_r$$

periodic B.C. $\phi_r(x, y, z) = \frac{1}{\sqrt{V}} e^{\frac{i}{\hbar} (P_x x + P_y y + P_z z)}$

$$-\frac{\hbar^2}{2m} \left[\left(\frac{i}{\hbar} P_x \right)^2 + \left(\frac{i}{\hbar} P_y \right)^2 + \left(\frac{i}{\hbar} P_z \right)^2 \right] = E_r$$

$$\rightarrow E_r = \frac{1}{2m} (P_x^2 + P_y^2 + P_z^2)$$

$$e^{\frac{i}{\hbar} P_x x} = e^{\frac{i}{\hbar} P_x (x+L)} \quad \text{periodic B.C.}$$

$$e^{\frac{i}{\hbar} P_x L} = 1$$

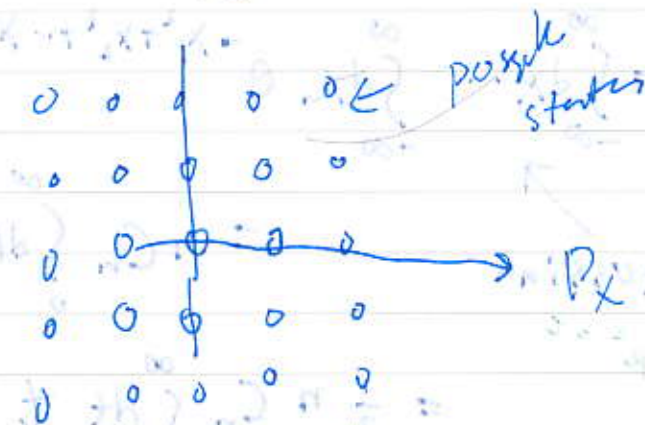
$$\frac{P_x L}{\hbar} = 2\pi m_x$$

$$m_x = 0, \pm 1, \pm 2, \dots$$

$$P_x = \frac{2\pi m_x \hbar}{L} = \frac{m_x h}{L}$$

$$P_y = \frac{m_y h}{L}$$

$$P_z = \frac{m_z h}{L}$$



quantum numbers

$$r = (m_x, m_y, m_z)$$

Many-Particle system

N identical particles

$$H = \sum_{j=1}^N H^{(j)} + \text{interaction}$$

E.g. electrons in a periodic lattice potential $V(x)$

$$H^{(j)} = -\frac{\hbar^2}{2m} \nabla_j^2 + V(\vec{x}_j)$$



Let $H^{(j)} \phi_r(j) = \epsilon_r \phi_r(j)$ $r = 0, 1, 2, 3, \dots$

set of eigen energy and eigenfunction of a single particle.

quantum system is described by a ray in Hilbert space

$$\vec{x} = (\vec{r}, \sigma)$$

full wave function

$$\Psi(1, 2, \dots, N) \equiv \Psi(\vec{x}_1, \vec{x}_2, \dots, \vec{x}_N)$$

$\Psi \rightarrow e^{i\theta} \Psi$ represent the same physical reality. in case there is spin

$\Psi \rightarrow \lambda \Psi$ is the same $\|\Psi\|=1$ fix λ up to a phase factor

$$\Psi(1, 2, \dots, i, \dots, j, \dots, N) = \alpha \Psi(1, 2, \dots, j, \dots, i, \dots, N)$$

↑
phase (indep of i, j)

$$= \alpha^2 \Psi(1, 2, \dots, i, \dots, i, \dots, N) \Rightarrow \alpha^2 = 1$$

$$\alpha = e^{i\phi} \quad \phi = 0, \pi$$

$$\alpha = \pm 1$$

$$\alpha = +1 \quad \text{bosons}$$

$$-1 \quad \text{fermions}$$

Talk about this general principle first.

$$\Psi_B(1, 2, \dots, N) = \sum_P \prod \phi_r(j) \leftarrow$$

↑
full permutation

each state r can be occupied by any number of particles $n_r = 0, 1, 2, \dots$

Pauli exclusion principle $\leftarrow n_r = 0, 1$ only apply

$$\Psi_F(1, 2, \dots, N) = \sum_P (-1)^P \prod \phi_r(j) \propto \text{Slater determinant}$$

grand-partition function for Boson/Fermion system
of identical particles non interacting

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Date

$$\Xi = \text{Tr} e^{-\beta(H - \mu N)}$$

occupation
number
representation
↓

$$= \sum_{\psi} \langle \psi | e^{-\beta(H - \mu N)} | \psi \rangle$$

$$|\psi\rangle = |n_0, n_1, n_2, \dots, n_r, \dots\rangle$$

$$\propto (a_0^\dagger)^{n_0} (a_1^\dagger)^{n_1} \dots |0\rangle$$

$$= \sum_{n_0, n_1, n_2, \dots, n_r, \dots} e^{-\beta \left(\sum_{r=0}^{\infty} \epsilon_r n_r - \mu \sum_{r=0}^{\infty} n_r \right)}$$

r specifies the
"states" not
energy

$$= \sum_{n_0, n_1, n_2, \dots, n_r, \dots} \prod_{r=0}^{\infty} e^{-\beta(\epsilon_r - \mu)n_r} = \prod_{r=0}^{\infty} \sum_{n_r} e^{-\beta(\epsilon_r - \mu)n_r}$$

$$\sum_{i,j,k,\dots} a_i \cdot b_j \cdot c_k \dots = (\sum_i a_i) (\sum_j b_j) (\sum_k c_k) \dots$$

$$= \prod_{r=0}^{\infty} \begin{cases} [1 + e^{-\beta(\epsilon_r - \mu)}] & \text{Fermions} \\ \frac{1}{1 - e^{-\beta(\epsilon_r - \mu)}} & \text{Bosons} \end{cases}$$

Fermions

Bosons

$$= \prod_{r=0}^{\infty} [1 + \eta e^{-\beta(\epsilon_r - \mu)}]^\eta$$

$$\eta = +1 \quad \text{Fermi} \\ -1 \quad \text{Boson}$$

grand potential

$$\Psi = U - TS - \mu N = -pV = -k_B T \ln \Xi$$

$$= -k_B T \sum_{r=0}^{\infty} \eta \ln(1 + \eta e^{-\beta(\epsilon_r - \mu)})$$

$$d\Psi = -S dT - p dV - \langle N \rangle d\mu$$

$$S = - \frac{\partial \Psi}{\partial T} = \frac{\partial}{\partial T} [k_B T \sum_{r=0}^{\infty} \eta \ln(1 + \eta e^{-\beta(\epsilon_r - \mu)})]$$

② $S = k_B \sum_r \eta \ln(1 + \eta e^{-\beta(\epsilon_r - \mu)}) + k_B T \sum_r \eta \cdot \frac{\eta e^{-\beta(\epsilon_r - \mu)} \frac{(\epsilon_r - \mu)}{k_B T^2}}{1 + \eta e^{-\beta(\epsilon_r - \mu)}}$

$$= k_B \sum_r \eta \ln(1 + \eta e^{-\beta(\epsilon_r - \mu)}) + k_B \sum_r \frac{\frac{\epsilon_r - \mu}{k_B T} e^{-\beta(\epsilon_r - \mu)}}{1 + \eta e^{-\beta(\epsilon_r - \mu)}}$$

① $\langle N \rangle = - \frac{\partial \Psi}{\partial \mu} = k_B T \sum_{r=0}^{\infty} \eta \frac{\eta e^{-\beta(\epsilon_r - \mu)}}{1 + \eta e^{-\beta(\epsilon_r - \mu)}} \beta$ + ferm

$$= \sum_{r=0}^{\infty} \frac{1}{e^{\beta(\epsilon_r - \mu)} + \eta} = \sum_r \langle n_r \rangle$$

$$\langle n_r \rangle = \frac{1}{e^{\beta(\epsilon_r - \mu)} + \eta}$$

↑
- boson

③ $U = \Psi + TS + \mu \langle N \rangle$

$$= -k_B T \sum_{r=0}^{\infty} \eta \ln(1 + \eta e^{-\beta(\epsilon_r - \mu)})$$

$$+ k_B T \sum_r \eta \ln(1 + \eta e^{-\beta(\epsilon_r - \mu)})$$

$$+ k_B T \sum_r \frac{(\epsilon_r - \mu) \eta e^{-\beta(\epsilon_r - \mu)}}{1 + \eta e^{-\beta(\epsilon_r - \mu)}} + \mu \sum_r \frac{1}{e^{\beta(\epsilon_r - \mu)} + \eta}$$

$$= \sum_r \frac{\epsilon_r}{e^{\beta(\epsilon_r - \mu)} + \eta} = \sum_{r=0}^{\infty} \epsilon_r \langle n_r \rangle$$

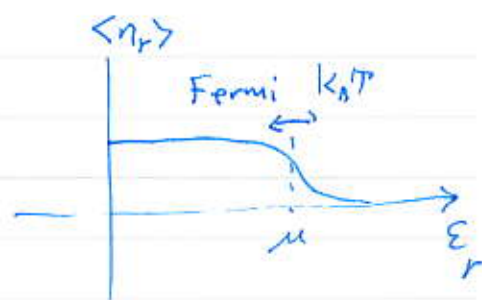
avg
occupati number

$$\langle n_r \rangle = \frac{1}{e^{\beta(\epsilon_r - \mu)} + \eta}$$

$\eta = +1$ fermi
 -1 boson

Eq. of state

$$PV = k_B T \sum_{r=0}^{\infty} \eta \ln(1 + \eta e^{-\beta(\epsilon_r - \mu)})$$



$$S = k_B \sum_r \left\{ \eta \left[-\beta(\epsilon_r - \mu) - \ln \langle n_r \rangle \right] + k_B \beta(\epsilon_r - \mu) \langle n_r \rangle \right\}$$

$$\begin{aligned} \ln(1 \oplus \eta \langle n_r \rangle) &= \ln \left(\frac{e^{\beta(\epsilon_r - \mu)} + \eta}{e^{\beta(\epsilon_r - \mu)} + 1} \right) \\ &= \ln \left(1 \oplus \frac{\eta}{e^{\beta(\epsilon_r - \mu)} + 1} \right) = \ln \left(\frac{e^{\beta(\epsilon_r - \mu)} + \eta}{e^{\beta(\epsilon_r - \mu)} + 1} \right) \\ &= \beta(\epsilon_r - \mu) + \ln \langle n_r \rangle \end{aligned}$$

$$\begin{aligned} &= \sum_R k_B \sum_r \left\{ \eta \left[-\ln(1 - \eta \langle n_r \rangle) + \ln \langle n_r \rangle - \ln \langle n_r \rangle \right] \right. \\ &\quad \left. + \left[\ln(1 - \eta \langle n_r \rangle) - \ln \langle n_r \rangle \right] \langle n_r \rangle \right\} \end{aligned}$$

$$S = k_B \sum_r \eta \ln \left[e^{-\beta(\epsilon_r - \mu)} (e^{\beta(\epsilon_r - \mu)} + \eta) \right] + k_B$$

$$k_B \beta(\epsilon_r - \mu) \frac{1}{e^{\beta(\epsilon_r - \mu)} + \eta}$$

$$= k_B \sum_r \left\{ -\eta \ln(1 - \eta \langle n_r \rangle) + \ln(1 - \eta \langle n_r \rangle) \frac{\langle n_r \rangle}{\langle n_r \rangle} - \ln \langle n_r \rangle \right\}$$

$$= k_B \sum_r \left\{ (\langle n_r \rangle - \eta) \ln(1 - \eta \langle n_r \rangle) - \langle n_r \rangle \ln \langle n_r \rangle \right\}$$

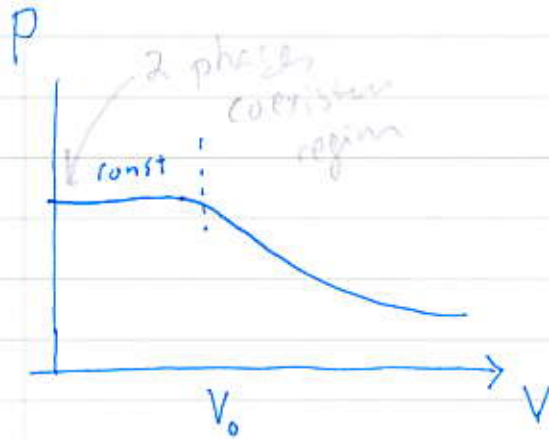
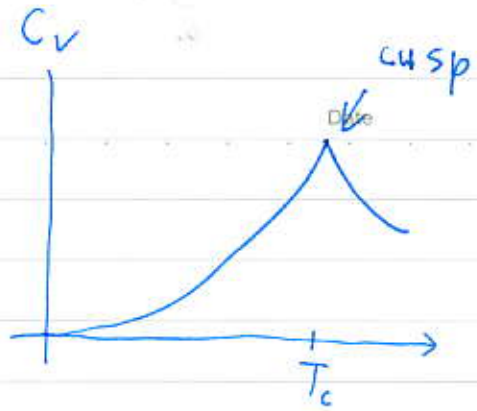
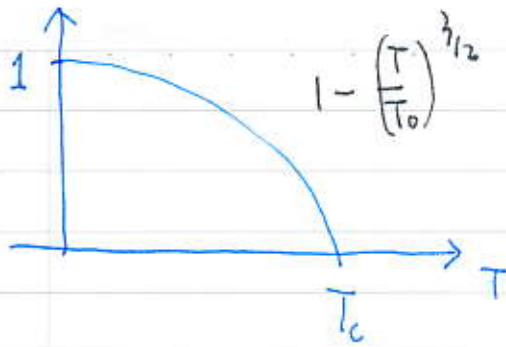
$\eta = +1$ for
 -1 boson

^4He was thought to be a realization of Bose-Einstein condensation system. but ^4He is not ideal quantum gas as ^4He has strong interaction.

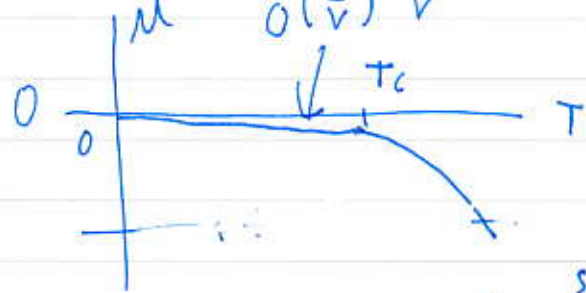
observed in dilute gas of Li ^{alkali atom} in 1995.

around microkelvin $\sim 10^{-6} \text{ K}$ or Sodium atoms

$$\langle n_0 \rangle / \langle N \rangle$$



E.g. ^4He $T_0 \approx 2\text{ K}$
 $\frac{N}{V} \approx 2 \times 10^{22} \text{ cm}^{-3}$



single particle

what is T_c ?

gap of energy levels $E_0 = 0$ $E_1 = \frac{p^2}{2m} \approx \frac{1}{m} \frac{h^2}{L^2}$

thermal energy $k_B T$ becomes compare to ΔE

$$k_B T_c \approx \frac{h^2}{m \lambda^2} \approx \frac{h^2}{m \left(\frac{V}{N}\right)^{2/3}}$$

λ
avg. spacing between particles

$$\frac{N}{V} = 2.612 \frac{(2\pi m k_B T_c)^{3/2}}{h^3} \quad \left(\frac{N}{V}\right)^{2/3} = (2.612)^{2/3} \frac{2\pi m k_B T_c}{h^2}$$

$$k_B T_c \approx \frac{(2.612)^{2/3}}{2\pi m} \frac{h^2}{\left(\frac{N}{V}\right)^{2/3}} \approx 0.3 \frac{h^2}{m \left(\frac{V}{N}\right)^{2/3}}$$

13 Feb 07

ch. 2. A. L. Fetter & J. D. Walecka
"Quantum Theory of many-particle systems" 95

convert discrete sum by integration

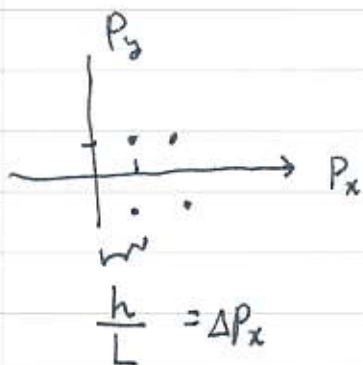
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$$\sum_r f_r = \sum_{\Delta p} f_{\epsilon(p)} (\Delta p) \left(\frac{L}{h}\right)^3 \stackrel{\text{momentum}}{=} \sum_p f_{\epsilon(p)} \cdot 1$$

$$\lim_{L \rightarrow \infty} = \frac{V}{h^3} \int f(\epsilon(p)) d^3p$$

$$r = (m_x, m_y, m_z)$$

$$m_x, m_y, m_z = 0, \pm 1, \pm 2, \dots$$



in terms of energy

$$\frac{p^2}{2m} = \epsilon$$

$$\frac{p dp}{m} = d\epsilon \quad dp = \frac{m d\epsilon}{p}$$

$$= \frac{m d\epsilon}{\sqrt{2m\epsilon}}$$

$\left(\frac{L}{h}\right)^3$: density of state per unit momentum volume

$$\sum_r f_r = \frac{V}{h^3} \int f(\epsilon) \cdot 4\pi \cdot 2m\epsilon \cdot \frac{m d\epsilon}{\sqrt{2m\epsilon}}$$

$$= \int \frac{V}{h^3} \cdot 4\pi \sqrt{2\epsilon} m^{\frac{3}{2}} d\epsilon f(\epsilon) d\epsilon$$

$$= \int_0^\infty D(\epsilon) f(\epsilon) d\epsilon = \frac{\sqrt{2\pi} (2m\epsilon)^{\frac{1}{2}}}{h^3}$$

$$D(\epsilon) = \frac{V}{h^3} 4\pi m^{\frac{3}{2}} \sqrt{2\epsilon} \quad \leftarrow \text{density of states in energy}$$

$$\Psi(T, V, \mu) = -k_B T \int_0^\infty d\epsilon \cdot D(\epsilon) \ln(1 + \eta e^{-\beta(\epsilon - \mu)}) = -pV$$

$$\eta = +1 \text{ fermion}$$

$$\eta = -1 \text{ boson}$$

$$\langle N \rangle = \int_0^\infty d\varepsilon \frac{D(\varepsilon)}{e^{\beta(\varepsilon-\mu)} + \eta}$$

$$D(\varepsilon) \propto \sqrt{\varepsilon}$$

$$\text{Let } D(\varepsilon) = a \sqrt{\varepsilon}$$

$$U = \int_0^\infty d\varepsilon \frac{\varepsilon D(\varepsilon)}{e^{\beta(\varepsilon-\mu)} + \eta}$$

$$pV = k_B T \int_0^\infty d\varepsilon D(\varepsilon) \cdot \eta \ln(1 + \eta e^{-\beta(\varepsilon-\mu)})$$

$$= \frac{1}{\beta} \eta \int_0^\infty \ln(1 + \eta e^{-\beta(\varepsilon-\mu)}) d(a \varepsilon^{\frac{3}{2}}) \frac{2}{3}$$

$$= \frac{2\eta}{3\beta} \left[a \varepsilon^{\frac{3}{2}} \ln(1 + \eta e^{-\beta(\varepsilon-\mu)}) \right]_0^\infty - \int_0^\infty \frac{a \varepsilon^{\frac{3}{2}} \cdot \eta e^{-\beta(\varepsilon-\mu)}}{1 + \eta e^{-\beta(\varepsilon-\mu)}} (-\beta) d\varepsilon$$

$$= \frac{2}{3} \int_0^\infty \frac{\varepsilon D(\varepsilon)}{e^{-\beta(\varepsilon-\mu)} + \eta} d\varepsilon = \frac{2}{3} U$$

(ideal gas)

$$U = \frac{3}{2} N k_B T$$

$$\begin{aligned} \varepsilon^{\frac{3}{2}} \ln(1 + \eta e^{-\beta\varepsilon}) & \begin{cases} \varepsilon \rightarrow 0 & 0 \cdot \ln 1 = 0 \\ \varepsilon \rightarrow \infty & \varepsilon^{\frac{3}{2}} e^{-\beta\varepsilon} \rightarrow 0 \end{cases} \\ \varepsilon^{\frac{3}{2}} \ln(1 - e^{-\beta\varepsilon})|_{\varepsilon \rightarrow 0} & \\ \varepsilon^{\frac{3}{2}} \ln(\beta\varepsilon) \rightarrow 0 & \quad \ln(1+x) \approx x \quad x \text{ small} \end{aligned}$$

Bose-Einstein condensation

The replacement $\sum_r f_r = \int f(\varepsilon) D(\varepsilon) d\varepsilon$ was wrong for $T < T_c$

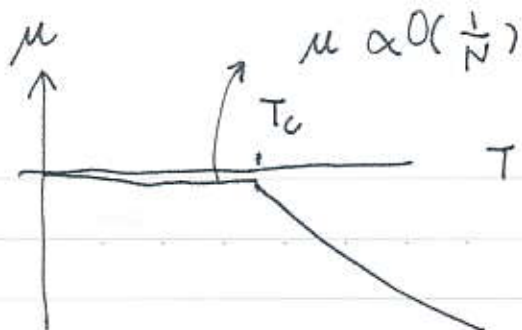
$$\sum_r f_r = \cancel{f_0} + \sum_{r \neq 0} f_r$$

$$= f_0 + \int f(\varepsilon) D(\varepsilon) d\varepsilon$$

$$\text{E.g. } \langle N \rangle = N_0 + \int_0^\infty d\varepsilon \frac{D(\varepsilon)}{e^{\beta(\varepsilon-\mu)} + \eta} \quad \text{for } T < T_c$$

$$\mu \rightarrow 0$$

how to determine T_c ?



T_c is determined by $\mu = 0$ 99

$$\langle N \rangle = \int_0^{\infty} d\varepsilon \frac{D(\varepsilon)}{e^{\frac{\varepsilon}{k_B T_c}} - 1}$$

$$\langle N \rangle = n_0 + \int_0^{\infty} d\varepsilon \frac{a \varepsilon^{1/2}}{e^{\beta \varepsilon} - 1} \quad \text{for } T < T_c \quad \beta \varepsilon = x$$

$$\varepsilon = \frac{x}{\beta} = k_B T x$$

$$= n_0 + \int_0^{\infty} \frac{(k_B T)^{3/2} a x^{1/2} dx}{e^x - 1} = n_0 + b T^{3/2}$$

$$n_0 = \langle N \rangle - b T^{3/2} = \langle N \rangle (1 - (T/T_c)^{3/2})$$

$$\langle N \rangle = \int_0^{\infty} d\varepsilon \frac{D(\varepsilon)}{e^{\beta(\varepsilon - \mu)} - 1}$$

$\varepsilon \geq 0$
 $\langle N \rangle > 0$
 if $\mu > 0$
 $e^{\beta(\varepsilon - \mu)} < 1$
 then $\langle N \rangle < 0$
 which is impossible

when $T \equiv 0$

$$\mu = \frac{E(N+1) - E(N)}{(N+1) - N} = 0$$

$$\varepsilon_0 = 0$$

$$\langle n_0 \rangle = \frac{1}{e^{\beta(\varepsilon_0 - \mu)} - 1} = \frac{1}{e^{-\beta \mu} - 1}$$

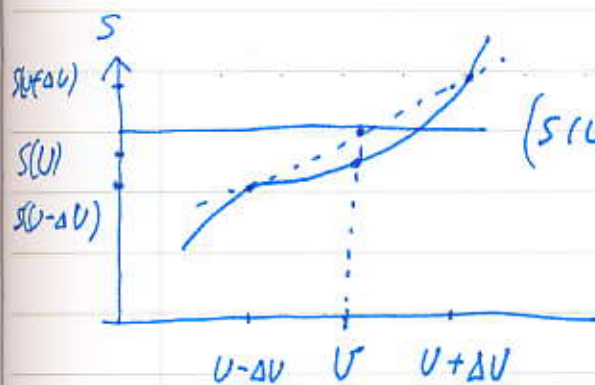
$\langle n_0 \rangle > 0$
 so $\mu \neq 0$

$$\text{assuming } \mu \text{ small} = \frac{1}{1 - \beta \mu + \dots - 1} = \frac{1}{\beta(-\mu)} \propto N$$

$$\text{so } \mu \propto \frac{1}{N}$$

$$= \frac{k_B T}{(-\mu)}$$

$$\mu \propto -\frac{k_B T}{N}$$



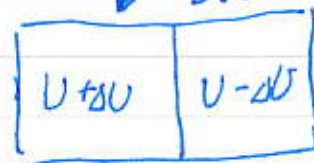
Date

original system



$$2S(U)$$

$$\checkmark S(U-\Delta U) + S(U+\Delta U)$$



if concavity
is violated,
the system
defies
inhomogeneity

concavity condition

$$2S(U) \geq S(U-\Delta U) + S(U+\Delta U) \leftarrow \text{global condition}$$

 $\leftarrow$ lies entropy

$$\text{or } S(U+\Delta U) - 2S(U) + S(U-\Delta U) \leq 0$$

it maximizes

not in concave

$$S(U+\Delta U) - S(U) - (S(U) - S(U-\Delta U)) \leq 0$$

again

$$S'(U)\Delta U - S'(U-\Delta U)\Delta U \leq 0$$

constrained

of partition the

entropy

$$S'(U) - S'(U-\Delta U)\Delta U \leq 0$$

$$S''(U)(\Delta U)^2 \leq 0$$

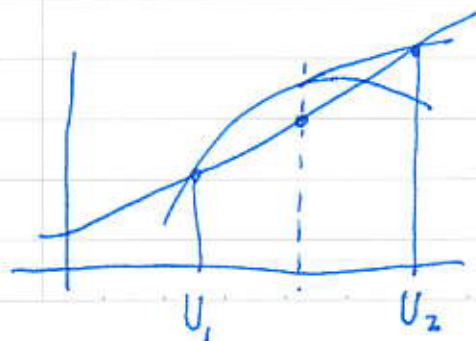
$$\rightarrow S''(U) \leq 0$$

$$\rightarrow \frac{\partial^2 S}{\partial U^2} \leq 0$$

local condition
less restrictive

$$\rightarrow \text{similarly } \frac{\partial^2 S}{\partial V^2} \leq 0$$

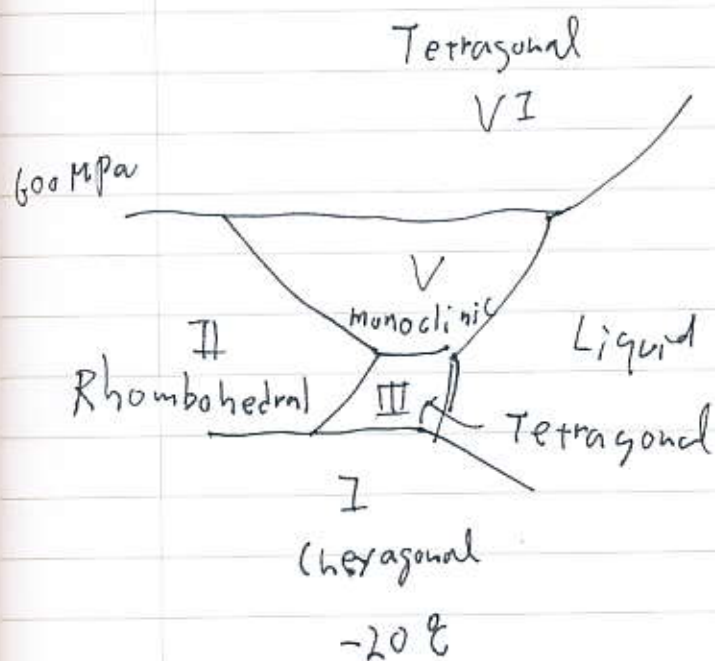
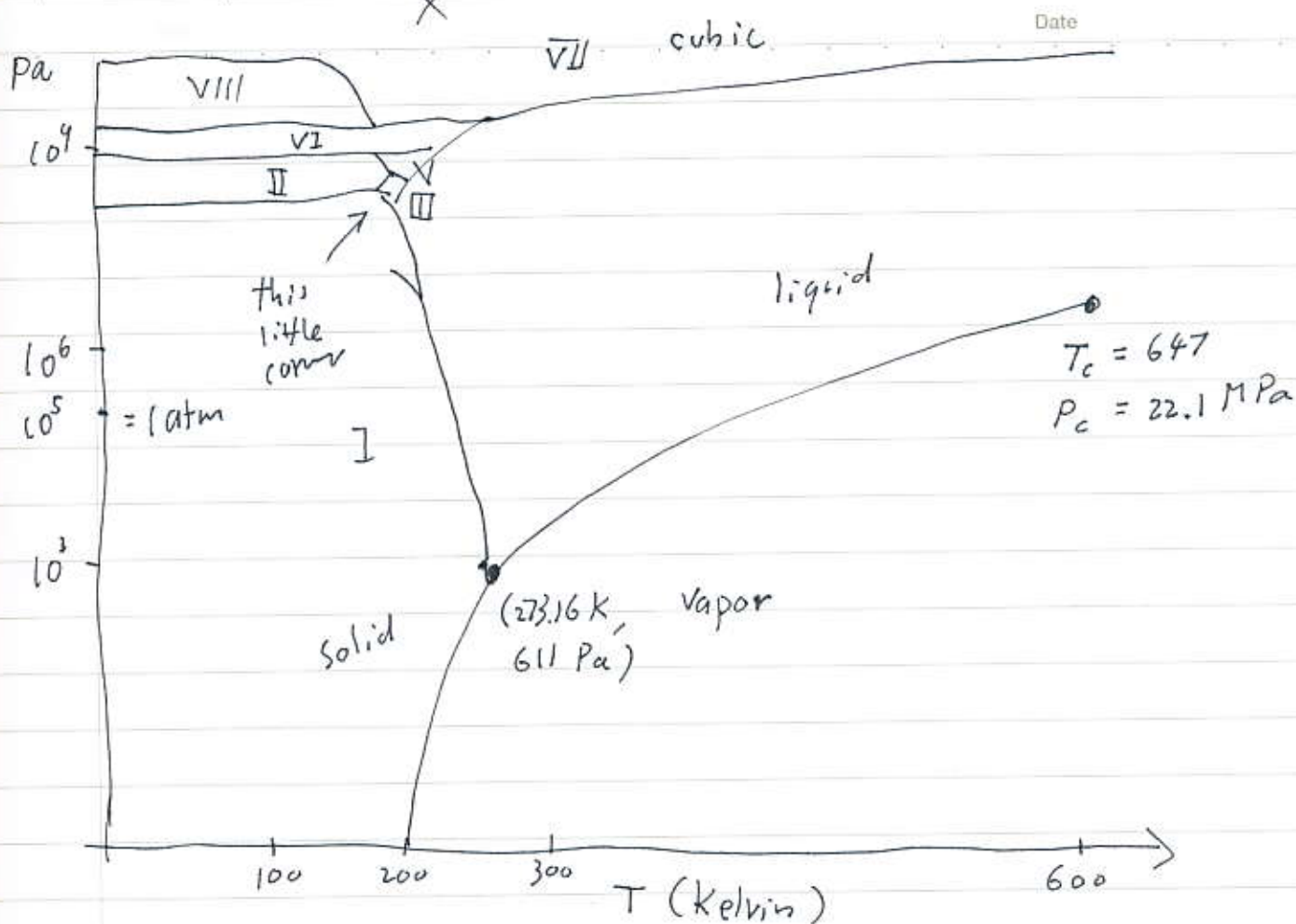
$$S(U+\Delta U, V+\Delta V, N) + S(U-\Delta U, V-\Delta V, N) \leq 2S(U, V, N)$$



$$\lambda S(U_1) + (1-\lambda)S(U_2) < S(\lambda U_1 + (1-\lambda)U_2)$$

$$0 \leq \lambda \leq 1$$

phase diagram of water XI



$$1 \text{ atm} = 101325 \text{ Pa}$$

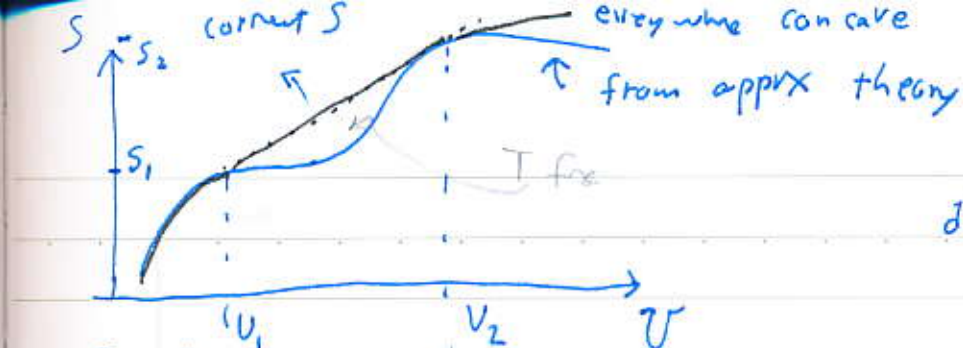
$$\approx 10^5 \text{ Pa}$$

$$1 \text{ Pa} = 1 \text{ kg}/(\text{m} \cdot \text{s}^2)$$

$$= 1 \text{ N}/(\text{m}^2)$$

snow?

pascal



$$dU = TdS + pdV + \mu dN$$

$$= \delta Q \quad dV=0 \quad dN=0$$

$$\Delta Q = T_0(S_2 - S_1)$$

↑ latent heat

going from states $T < T_0$

to states $T > T_0$

energy & entropy
jump discontinuously



1st order phase transition

consequence of the stability condition

$$\frac{\partial^2 S}{\partial U^2} \leq 0 \quad \rightarrow \quad \frac{\partial}{\partial U} \left(\frac{\partial S}{\partial U} \right)_{V,N} = \frac{\partial}{\partial U} \left(\frac{1}{T} \right)_{V,N} = -\frac{1}{T^2} \left(\frac{\partial T}{\partial U} \right)_{V,N}$$

$$= -\frac{1}{T^2} \frac{1}{C_V} < 0 \quad \rightarrow \quad C_V \geq 0$$

$$\frac{\partial^2 S}{\partial V^2} \leq 0 \quad \rightarrow \quad \frac{\partial}{\partial V} \left(\frac{\partial S}{\partial V} \right)_{U,N} = \frac{\partial}{\partial V} \left(\frac{P}{T} \right)_{U,N} ?$$

stability in other fundamental relation

$$S: \text{concave in } U, V \quad \frac{\partial^2 S}{\partial U^2} \leq 0 \quad \frac{\partial^2 S}{\partial V^2} \leq 0$$

Date

$$U(S, V, N) \quad \text{convex in } S, \quad \text{con}^{\text{vex}} \text{ in } V$$

$$\frac{\partial^2 U}{\partial S^2} \geq 0 \quad \frac{\partial^2 U}{\partial V^2} \geq 0$$

$$F(T, V, N) \quad \text{concave in } T$$

$$\frac{\partial^2 F}{\partial T^2} \leq 0 \quad \frac{\partial^2 F}{\partial V^2} \geq 0$$

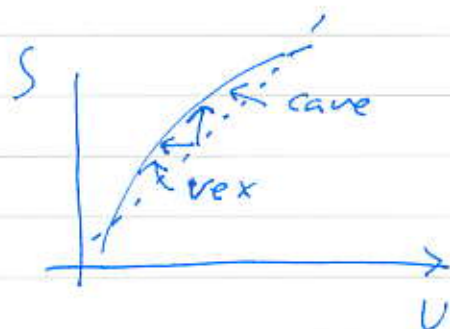
$$G(T, P, N) = F - \mu N \quad \text{concave in } T \text{ \& } P$$

$$\frac{\partial^2 G}{\partial T^2} \leq 0 \quad \frac{\partial^2 G}{\partial P^2} \leq 0$$

proof: from $\frac{\partial^2 S}{\partial U^2} \leq 0 \quad \frac{\partial^2 S}{\partial V^2} \leq 0$ to condition for U

$$F = U - TS$$

$$T = \left. \frac{\partial U}{\partial S} \right|_{V, N} \quad S = - \left. \frac{\partial F}{\partial T} \right|_{V, N}$$

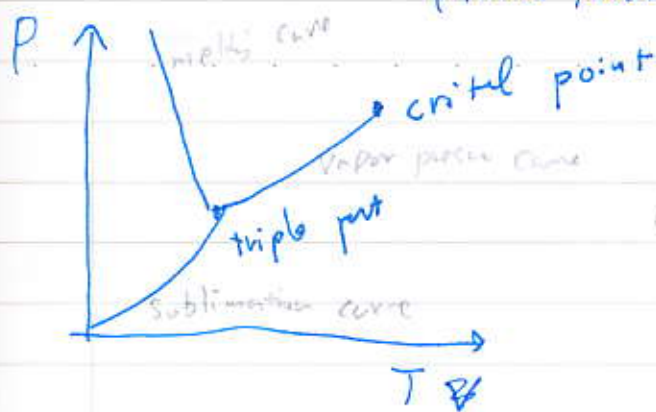


$$\left. \frac{\partial S}{\partial T} \right|_{V, N} = - \left. \frac{\partial^2 F}{\partial T^2} \right|_{V, N}$$

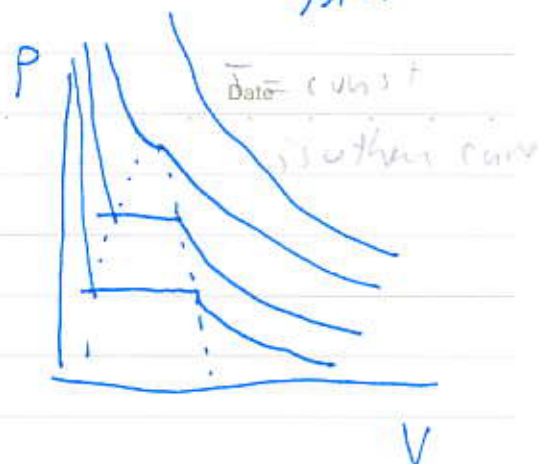
$$= \left(\frac{\partial T}{\partial S} \right)^{-1} \bigg|_{V, N} = \left. \frac{\partial^2 U}{\partial S^2} \right|_{V, N}$$

qualitative features of phase diagrams in simple one-component system

(Thomas Andrews 1869)



Ar
(Argon)



Clapeyron eqn.

$$\frac{dP}{dT} = \frac{l}{T \Delta v} \leftarrow \text{latent heat per mole} \leftarrow \text{volume change per mole}$$

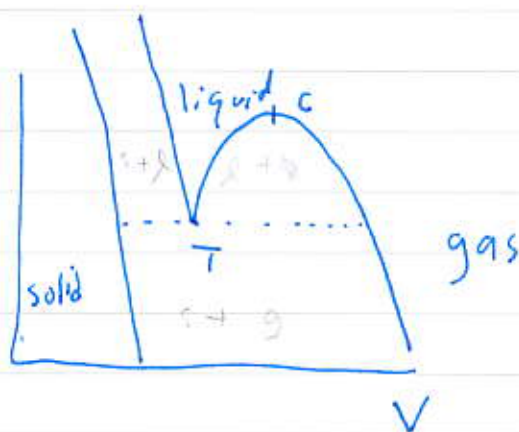
$$l = T \Delta s$$

See Callen page 229-230

solid	liquid
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conditions for phase equilibrium

$$\begin{aligned} T_1 &= T_2 \\ P_1 &= P_2 \\ \mu_1 &= \mu_2 \end{aligned}$$

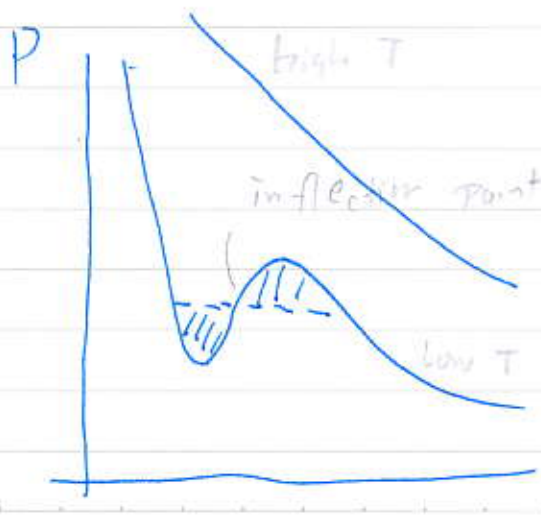


van der Waals invented the law of corresponding states

van der Waals theory of phase trans. is

$$\left[P + a \left(\frac{N}{V} \right)^2 \right] (V - Nb) = N k_B T$$

year (1873)
nobel prize in 1910



Maxwell constructs by requiring $\mu_1 = \mu_2$

derivation of van der Waals equation

van der Waals theory of phase transition 1873.

Ideal gas

$$F = -k_B T \ln Z$$

$$Z = \int d\mathbf{p}_1 \dots d\mathbf{p}_{3N} \int d\mathbf{q}_1 \dots d\mathbf{q}_{3N} e^{-\frac{\sum \mathbf{p}_i^2}{2m}} \frac{1}{N! h^{3N}}$$

$$= V^N \cdot T^{\frac{3N}{2}} \cdot \text{const}$$

$$F = -k_B T N \ln V + f(T) = U(T) - TS$$

$$P = -\frac{\partial F}{\partial V} = \frac{k_B T N}{V}$$

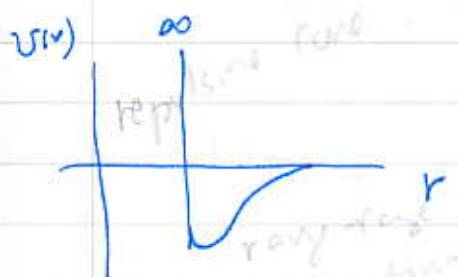
$$S = -\frac{\partial F}{\partial T} = k_B N \ln V + \text{terms indep of } V$$

$V(r)$

Lennard-Jones potential

$$V(r) = \epsilon \left(\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right)$$

r = distance between two particles



when $V \rightarrow \infty$

$$e^{-\infty} = 0$$

$$Z = \int d\mathbf{p}_1 \dots d\mathbf{p}_{3N} e^{-\beta \sum_{ij} V(r_{ij})} \cdot T^{\frac{3N}{2}}$$

$$\vec{r}_1 = (q_1, q_2, q_3)$$

$$\vec{r}_2 = (q_4, q_5, q_6)$$

$$\int d\vec{r} e^{-\beta \dots} < V$$

mean-field approx.



other particles are fixed at their equlib positions
look at one fixed particle say $\vec{r}_1$

$$\int d\vec{r} e^{-\beta \sum V(r_{ij})} = V - bN$$

excluded volume effect

$$\rightarrow S = k_B N \ln(V - bN) + \text{terms indep of } V$$

Volume dependent contribution to internal energy U

$$U = -a \left(\frac{N}{V} \right) \cdot N + V \text{ indep't part.}$$

↑
attractive

↑
every particle
contribution

$\frac{N}{V} = \rho$ is density



range of attract

if there is a particle there
will be contribution

$$\sum_{i < j} v(\vec{r}_{ij}) \quad \vec{r}_{ij} \text{ random}$$

$$U = \sum_{i < j} \langle \epsilon v \rangle = N(N-1) \langle v \rangle$$

$$F = U - TS$$

$$\langle v \rangle = \frac{\sigma^3}{V} \epsilon_0 + 0$$

$\rho \cdot \sigma^3$ is total contribution
to each particle

$$= -a \left(\frac{N^2}{V} \right) - Nk_B T \ln(V - Nb) + \text{terms indep't of volume.}$$

terms indep't of volume has no contribution

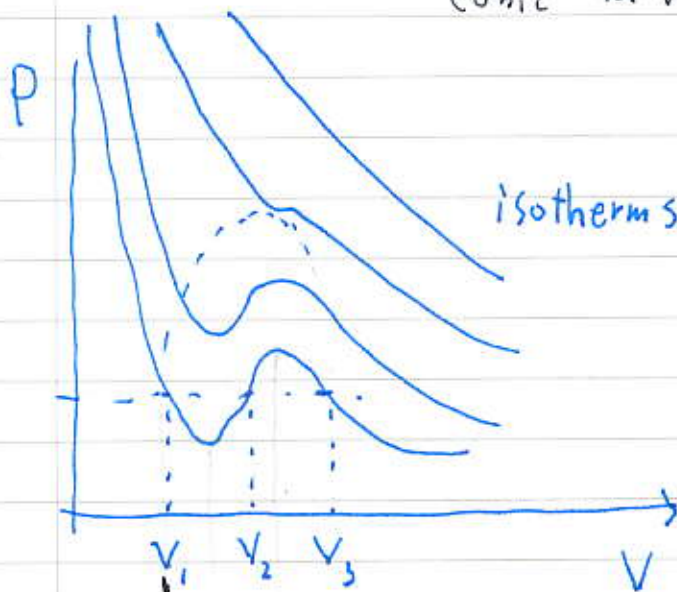
$$p = - \frac{\partial F}{\partial V} = -a \frac{N^2}{V^2} + \frac{Nk_B T}{V - Nb}$$

$$U \propto \frac{N^2}{V}$$

$$\left[p + a \left(\frac{N}{V} \right)^2 \right] (V - Nb) = Nk_B T$$

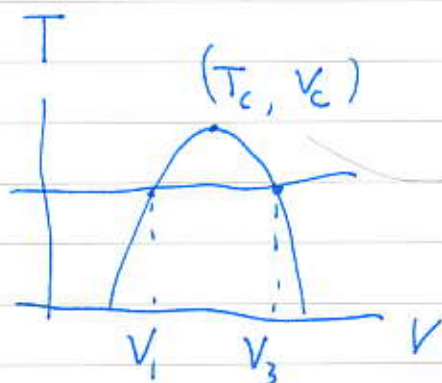
cubic in V

critical point



$$\left. \frac{\partial p}{\partial V} \right|_T = 0 \quad \text{two extremes}$$

$$\frac{\partial^2 p}{\partial V^2} = 0 \quad \text{point of inflection}$$

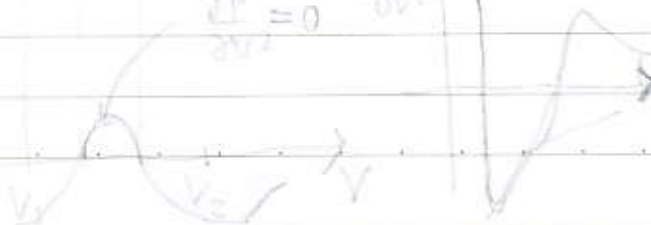


$$\frac{\partial p}{\partial V}$$

$$\frac{\partial^2 p}{\partial V^2} = 0$$

$$\frac{\partial^2 p}{\partial V^2}$$

0



$$\frac{\partial P}{\partial V} = \frac{2aN^2}{V^3} + \frac{(-NK_bT)}{(V-Nb)^2} = 0 \quad (1)$$

$$\frac{\partial^2 P}{\partial V^2} = -\frac{6aN^2}{V^4} + \frac{2NK_bT}{(V-Nb)^3} = 0 \quad (2)$$

$$\frac{(1)}{(2)} \quad \frac{2}{3}V = V - Nb, \quad bN = \frac{1}{3}V$$

$$V_c = 3bN, \quad \frac{2aN^2}{(3bN)^3} = \frac{NK_bT_c}{\left[\frac{2}{3}(3bN)\right]^2}$$

$$K_bT_c = \frac{8}{27} \frac{a}{b}$$

Substitute into (1)

$$\frac{2a}{3^3 \cdot b} \left(\frac{2}{3}\right)^2 = K_bT_c$$

$$P_c = -a\left(\frac{N}{V_c}\right)^2 + \frac{NK_bT_c}{V_c - Nb} = -a\frac{1}{(3b)^2} + \frac{8}{27} \frac{a}{b} \frac{1}{2b} = \frac{a}{27b^2}$$

Maxwell construction

Gibbs-Duhem relation

$$U(\lambda S, \lambda V, \lambda N) = \lambda U(S, V, N)$$

→ $\frac{\partial}{\partial \lambda}$ on both side, let $\lambda = 1$

$$U = TS - PV + \mu N$$

$$dU = TdS - PdV + \mu dN \Rightarrow SdT - Vdp + Nd\mu = 0$$

$$d\mu = -\frac{S}{N}dT + \frac{V}{N}dP$$

on isotherm curve $dT = 0$
 $T = \text{const}$

$$d\mu = \frac{V}{N}dP$$

$$\int_1^3 d\mu = \frac{1}{N} \int_1^3 V dP = \mu(3) - \mu(1) = 0$$

$$① \quad N = N_g + N_l \quad (\rho V = N)$$

$$\text{or } N_g = \frac{N}{V_g} V_g^x$$

$$N_l = \frac{N}{V_l} V_l^x$$

$$② \quad v_g N_g + v_l N_l = V^x = V$$

$$N_g + N_l = N$$

$$v_g N_g + v_l (N - N_g) = V$$

$$(v_g - v_l) N_g = V - V_l$$

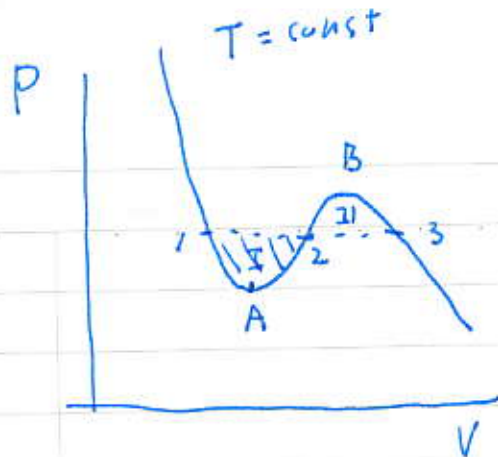
$$N_g = \frac{V - V_l}{v_g - v_l}$$

$$x_g = \frac{N_g}{N} = \frac{V - V_l}{V_g - V_l}$$

$$N_l = \frac{V_g - V}{v_g - v_l}$$

$$x_l = \frac{N_l}{N} = \frac{V_g - V}{V_g - V_l}$$

$$v_g N = V$$

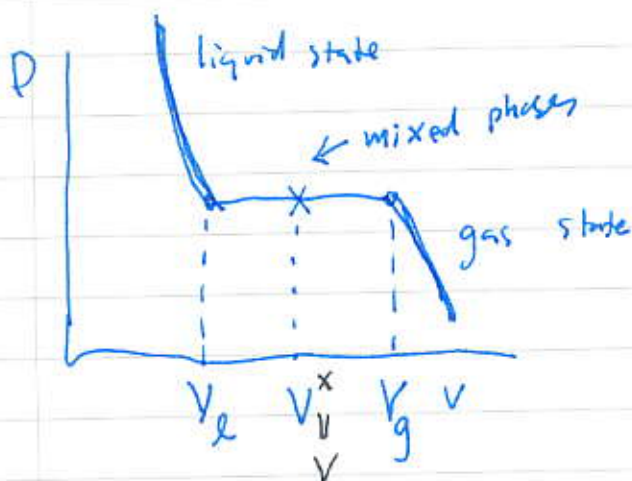


$$\int_1^3 v dp = \int_1^A + \int_A^2 + \int_2^B + \int_B^3 = 0 \quad (117)$$

$$= \int_A^2 - \int_A^1 + \int_2^B - \int_B^3$$

$$= \text{Area I} - \text{Area II} = 0$$

areas are equal.



same P
same T
same μ

$$V_g^x + V_l^x = V^x$$

see Callen ch.9.

The fraction of the system that exists in each of the two phases is governed by the "lever rule".

Let $\left\{ \begin{array}{l} v_l^x = \frac{V_l^x}{N} \\ v_g^x = \frac{V_g^x}{N} \end{array} \right.$ (volume per mole)

$$x_l = \frac{N_l}{N} \quad x_g = \frac{N_g}{N}$$

$$V^x = V_g^x + V_l^x = \frac{V_g^x}{N_g} N_g + \frac{V_l^x}{N_l} N_l = v_g N_g + v_l N_l$$

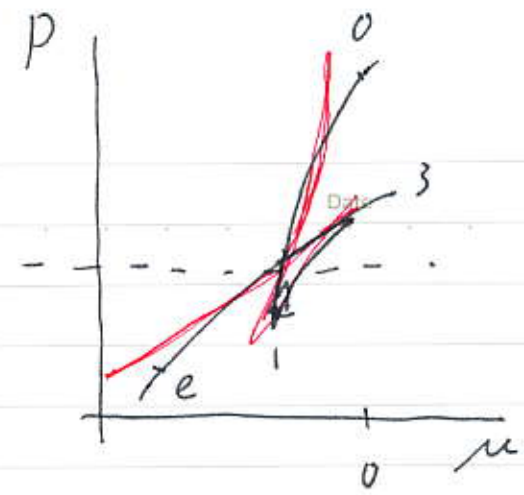
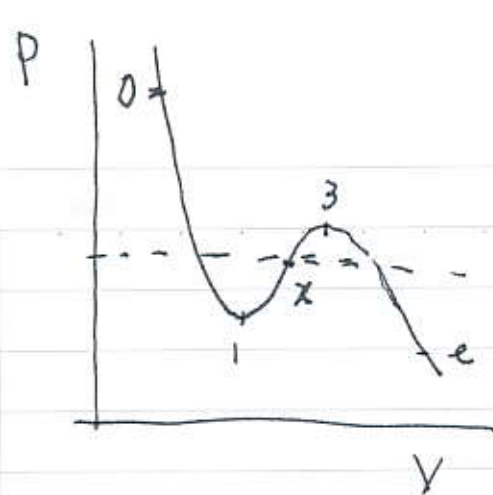
$$① N = N_g + N_l$$

$$② V_g^x + V_l^x = V^x$$

$$③ V_g^x = \frac{V_g^x}{N} N_g = v_g N_g$$

$$④ V_l^x = \frac{V_l^x}{N} N_l = v_l N_l$$

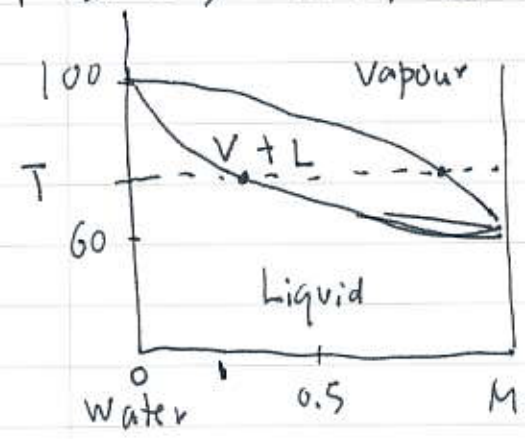
$$\left\{ \begin{array}{l} V_g^x + V_l^x = V^x \end{array} \right.$$



$$\mu = \mu_0 + \int_0^x \frac{v}{N} dP = \mu_0 + \int_0^x v dP = \frac{G}{N}$$

minimization principle in G or μ at fixed T, P, N
the system takes one that has smallest μ .

phase diagrams of binary systems



there are $T, P, \mu_1, \mu_2, \dots, \mu_r$
2 + r intensive parameters
M phases in each phase then are
1 Gibbs-Duhem relation so
total free intensive
vars is $r + 2 - M$

Gibbs phase rule

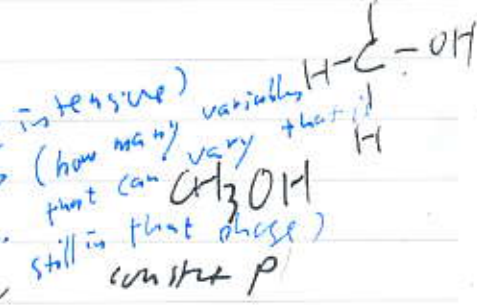
$$f = r - M + 2$$

r: # of components

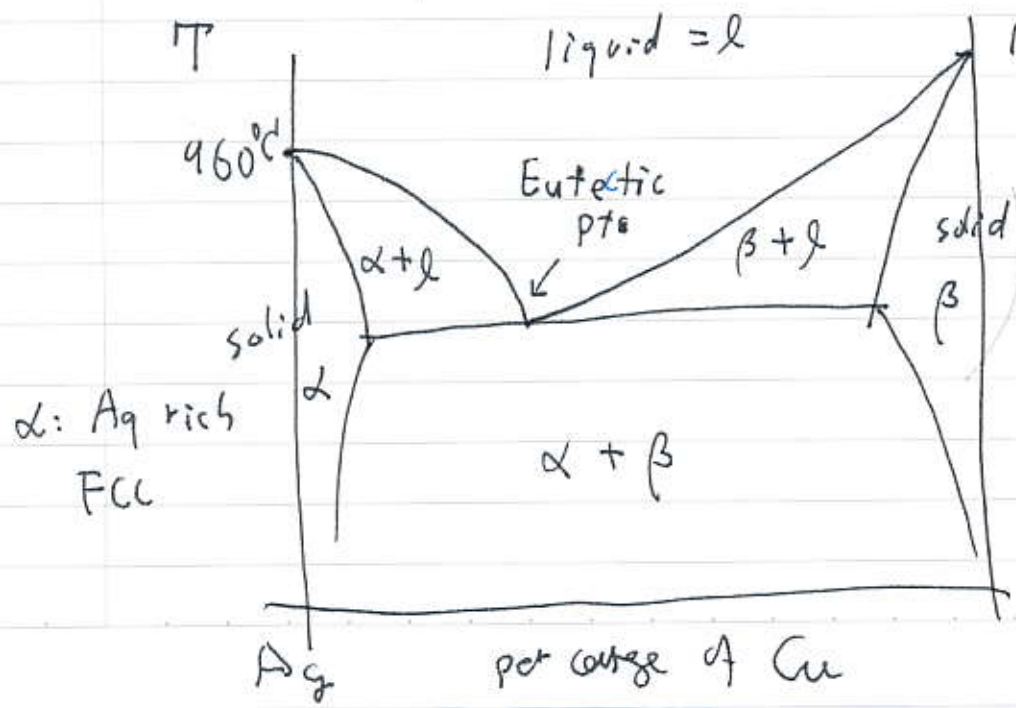
M: # of phases (intensive)

f: number of variables (how many vary that can vary)

still can vary



Mole fraction of Methanol 酒精



β Cu-rich FCC

per cent of Cu

$$q(t) = A \cos \omega t + B \sin \omega t$$

$$B = \frac{p}{m\omega}$$

$$q(0) = A$$

Date

$$\dot{q}(0) = A(-\omega) \sin \omega t + B \cos \omega t \cdot \omega = \frac{p}{m}$$

$$q(t) = q \cos \omega t + \frac{p}{m\omega} \sin \omega t$$

3.

$$Z = \frac{1}{N! h^{3N}} \int dP_1 \cdots dP_{3N} e^{-\beta \sum \frac{P_i^2}{2m}} \int q_1 \cdots q_{3N}$$

$$= \frac{1}{N! h^{3N}} V^N \left[\int_{-\infty}^{\infty} dp e^{-\frac{\beta p^2}{2m}} \right]^{3N}$$

$$= \frac{1}{N! h^{3N}} V^N \left(m 2\pi k_B T \right)^{\frac{3N}{2}} = \frac{1}{N!} \left[\frac{V (2\pi m k_B T)}{h^3} \right]^{\frac{3}{2} N}$$

$$F = -k_B T \ln Z = \frac{N^N}{N! \lambda^{3N}}$$

$$= -k_B T \left\{ N \ln \left[\frac{V (2\pi m k_B T)}{h^3} \right]^{\frac{3}{2}} - N \ln N + N \right\}$$

$$= -N k_B T \left\{ \ln \left(\frac{V}{N} \frac{(2\pi m k_B T)^{\frac{3}{2}}}{h^3} \right) + 1 \right\}$$

$$S = -\frac{\partial F}{\partial T} = N k_B \left\{ \ln \left(\frac{V}{N \lambda^3} \right) + 1 \right\} + N k_B \frac{3}{2} \frac{1}{T} \quad \lambda = \frac{h}{\sqrt{2\pi m k_B T}}$$

$$= N k_B \left\{ \frac{5}{2} + \ln \left(\frac{V}{N \lambda^3} \right) \right\}$$

$$P = -\frac{\partial F}{\partial V} = \frac{N k_B T}{V}$$

$$N = \frac{\partial F}{\partial N} = -k_B T \left\{ \ln \frac{V}{N \lambda^3} + 1 \right\} + k_B T = -k_B T \ln \frac{V}{N \lambda^3}$$

$$U = F + TS = -Nk_B T \left\{ \ln \frac{V}{N\lambda^3} + 1 \right\}$$

123

$$+ Nk_B T \left\{ \frac{5}{2} + \ln \frac{V}{N\lambda^3} \right\}_{\text{Date}} = \frac{3}{2} Nk_B T$$

grand canonical

$$\Xi = \sum_{N=0}^{\infty} Z_N e^{\beta \mu N} = \sum_{N=0}^{\infty} \frac{1}{N!} \left(\frac{V}{\lambda^3} \right)^N e^{\beta \mu N}$$

$$= \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{V}{\lambda^3} e^{\beta \mu} \right)^N = e^{\frac{V}{\lambda^3} e^{\beta \mu}}$$

$$\Psi(T, V, \mu) = -k_B T \ln \Xi = -k_B T \frac{V}{\lambda^3} e^{\beta \mu} \quad \lambda \propto T^{-\frac{1}{2}}$$

$$pV = k_B T \left(\frac{V}{\lambda^3} e^{\beta \mu} \right) = k_B T N \quad = -k_B T N$$

$$\left[\frac{(2\pi m k_B T)^{3/2}}{h^3} \right]^{-1} = \frac{3}{2} \frac{(2\pi m k_B)^{1/2}}{h^3} T^{1/2}$$

$$S = - \frac{\partial \Psi}{\partial T} = \left[-k_B \frac{V}{\lambda^3} e^{\beta \mu} + k_B T \frac{V}{\lambda^3} e^{\beta \mu} \frac{\mu}{k_B T^2} - k_B T \frac{V}{\lambda^3} e^{\beta \mu} \frac{3}{2} \frac{1}{T} \right]$$

$$= + \frac{5}{2} k_B \frac{V}{\lambda^3} e^{\beta \mu} \ominus \left(\frac{\mu}{T} \right) \frac{V}{\lambda^3} e^{\beta \mu}$$

$$N = \frac{\partial \Psi}{\partial \mu} = k_B T \frac{V}{\lambda^3} e^{\beta \mu} \cdot \beta = \frac{V}{\lambda^3} e^{\beta \mu}$$

$$\rightarrow + \frac{5}{2} k_B N \ominus \frac{\mu}{T} N$$

$$\Psi = U - TS - \mu N$$

$$p = - \frac{\partial \Psi}{\partial V} = k_B T \frac{e^{\beta \mu}}{\lambda^3} = k_B T \frac{N}{V}$$

$$U = \Psi + TS + \mu N = -k_B T N + \frac{5}{2} k_B T N \ominus \mu N + \mu N = \frac{3}{2} k_B T N$$

Total 3.

$Z = \frac{1}{N!} \prod_{i=1}^N$ partition function for 1 molecule

125

$$Z_{trans} = \int \frac{d\vec{r}}{h^3} d\vec{p} e^{-\frac{\beta p^2}{2m}} = V \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2}$$

$$Z = Z_{trans} Z_{rot} \cdot Z_{vib}$$

A B

$$\frac{2\pi m k_B T}{h}$$

$$Z_{rot} = \sum_{j=0,1,2,\dots} e^{-\frac{\beta \hbar^2 (j+1)j}{2I}} \cdot (2j+1)$$

↑
due to degeneracy

$$Z_{vib} = \sum_{n=0,1,2,\dots} e^{-\beta \hbar \omega n} = \frac{1}{1 - e^{-\beta \hbar \omega}}$$

$$F = -k_B T N (\ln Z_{trans} + \ln Z_{rot} + \ln Z_{vib}) - \frac{\partial}{\partial \beta}$$

$$\frac{U}{N} = -\frac{\partial}{\partial \beta} (\ln Z_{trans} + \ln Z_{rot} + \ln Z_{vib}) = -\frac{\partial}{\partial \beta} \ln Z_{trans} - \frac{\partial}{\partial \beta} \ln Z_{rot} - \frac{\partial}{\partial \beta} \ln Z_{vib}$$

$$= -\frac{\partial}{\partial \beta} \left(\left(\frac{3}{2} \right) \ln \beta \right) + \ln \sum_j (2j+1) e^{-\frac{\beta \hbar^2 (j+1)j}{2I}} - \ln (1 - e^{-\beta \hbar \omega})$$

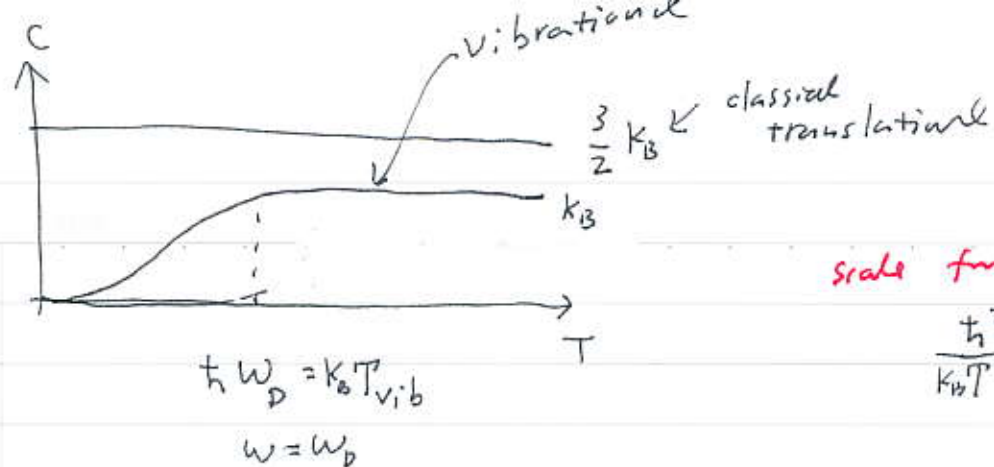
$$= \frac{3}{2} \frac{1}{\beta} + \frac{1}{Z_{rot}} \sum_j (2j+1) \left(\frac{\hbar^2 (j+1)j}{2I} \right) e^{-\frac{\beta \hbar^2 (j+1)j}{2I}} + \frac{e^{-\beta \hbar \omega} \hbar \omega}{1 - e^{-\beta \hbar \omega}}$$

$$= \frac{3}{2} k_B T + \frac{1}{Z_{rot}} \sum_j \frac{\hbar^2 (j+1)j}{2I} \cdot (2j+1) e^{-\frac{\beta \hbar^2 (j+1)j}{2I}} + \frac{\hbar \omega}{e^{\beta \hbar \omega} - 1}$$

$$\frac{C_v}{N} = \frac{d}{dT} \left(\frac{U}{N} \right) = \frac{3}{2} k_B + \frac{\hbar \omega e^{\beta \hbar \omega}}{(e^{\beta \hbar \omega} - 1)^2} \frac{\hbar \omega}{k_B T^2} + \dots$$

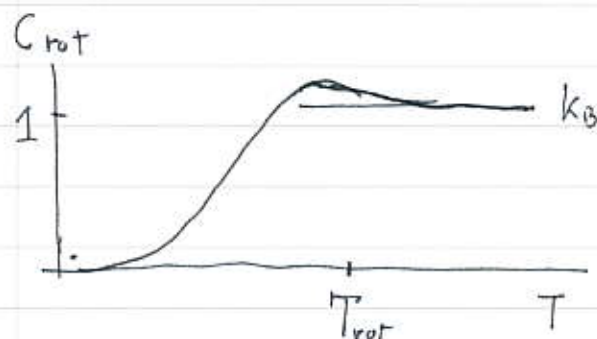
$$+ \frac{1}{Z_{rot}} \sum_j \left(\frac{\hbar^2 (j+1)j}{2I} \right)^2 (2j+1) e^{-\frac{\beta \hbar^2 (j+1)j}{2I}} \frac{1}{k_B T^2}$$

$$- \left(\frac{1}{Z_{rot}} \left[\sum_j \frac{\hbar^2 (j+1)j}{2I} (2j+1) e^{-\frac{\beta \hbar^2 (j+1)j}{2I}} \right] \right)^2 \frac{1}{k_B T^2}$$



Date:
 scale for rotational part

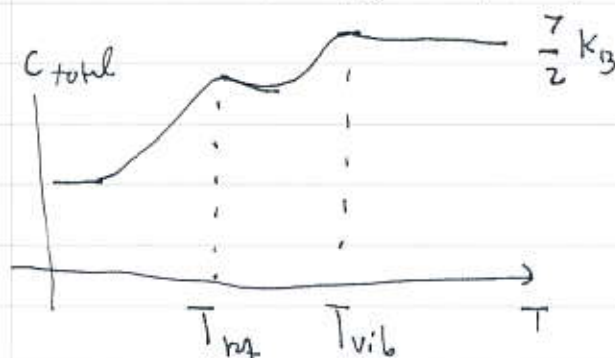
$$\frac{\hbar^2}{k_B T I} = O(1)$$



high T limit
$$Z_{rot} \approx \int_0^\infty dj e^{-\left(\frac{\hbar^2 j(j+1)}{2I}\right)} = \frac{2I}{\beta \hbar^2}$$

$$U = -\frac{\partial \ln Z_{rot}}{\partial \beta} = -\frac{\partial (-\ln \beta)}{\partial \beta} = \frac{1}{\beta} = k_B T$$

$$C = k_B \quad (\text{as } T \rightarrow \infty)$$



van der Waals Eq. behavior near the critical point

Date

$$\left[p + a \left(\frac{N}{V} \right)^2 \right] (V - bN) = Nk_B T$$

consider $p = p_c + \Delta p$

$$T = T_c + \Delta T$$

$$p_c = \frac{a}{27b^2}, \quad k_B T_c = \frac{8}{27} \frac{a}{b}, \quad V_c = 3bN \quad V = V_c + \Delta V$$

~~$$\left[p_c + \Delta p + a \frac{N^2}{V} \right]$$~~

~~$$\left[\frac{a}{27b^2} + \Delta p + a \frac{N^2}{(3bN + \Delta V)^2} \right] [3bN + \Delta V - bN] = Nk_B \Delta T + N \frac{8}{27} \frac{a}{b}$$~~

~~$$\left[\left(\frac{a}{27b^2} + \Delta p \right) (3bN + \Delta V)^2 + aN^2 \right] [2bN + \Delta V] + \left(Nk_B \Delta T + N \frac{8}{27} \frac{a}{b} \right) (3bN + \Delta V)^2$$~~

when $\Delta T = 0$

~~$$\left[\left(\frac{a}{27b^2} + \Delta p \right) (9b^2N^2 + 6bN \cdot \Delta V + \Delta V^2) + aN^2 \right] [2bN + \Delta V] = \frac{8}{27} N \frac{a}{b} (9b^2N^2 + 6bN \cdot \Delta V + \Delta V^2)$$~~

~~$$\left[\Delta p + aN^2 \left(-2 \frac{\Delta V}{V_c^3} \right) \right] (V_c - bN) + \left[p_c + a \left(\frac{N}{V_c} \right)^2 \right] \Delta V = Nk_B \Delta T$$~~

~~$$\left[\Delta p - \frac{2aN^2 \Delta V}{(3bN)^3} \right] (2bN) + \left[\frac{a}{27b^2} + \frac{a}{27b^2} \right] \Delta V = Nk_B \Delta T$$~~

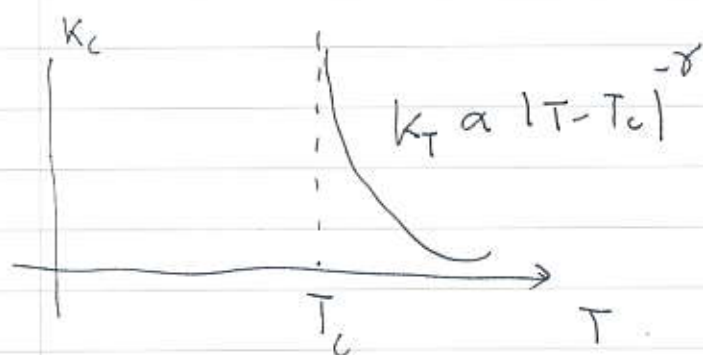
isothermal compressibility

$$K_T = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T$$

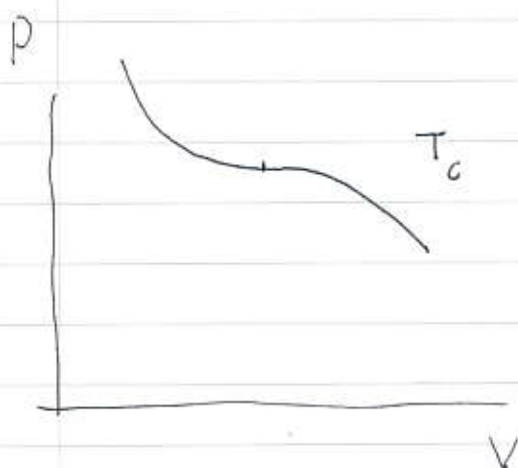
$$\left. \frac{\partial P}{\partial V} \right|_T = \frac{2aN^2}{V^3} - \frac{Nk_B T}{(V-Nb)^2}$$

$$\text{at } V=V_c \left\{ \begin{aligned} &= \frac{2aN^2}{27b^3N^3} - \frac{Nk_B T}{4N^2b^2} \end{aligned} \right.$$

$$\begin{aligned} K_T &= -\frac{1}{3bN} \frac{k_B(T_c - T)}{4N} \frac{4Nb^2}{k_B(T_c - T)} = \frac{2a}{27b^3N} - \frac{k_B T}{4Nb^2} \\ &= \frac{4}{3}b \frac{1}{k_B(T - T_c)} = \frac{1}{4} \left(\frac{8a}{27b} \right) \frac{1}{b^2N} - \dots \\ &= \frac{k_B(T_c - T)}{4Nb^2} \end{aligned}$$



$$\gamma = 1$$

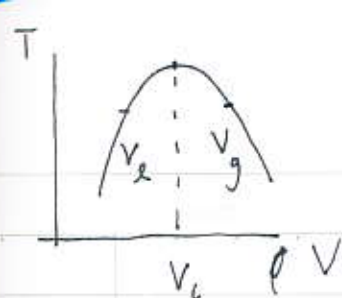


critical isotherm curve

$$P(V, T_c) = P_c + \left. \frac{\partial P}{\partial V} \right|_{V_c, T_c} (V - V_c) + \frac{1}{2} \left. \frac{\partial^2 P}{\partial V^2} \right|_{V_c, T_c} (V - V_c)^2 + \frac{1}{6} \left. \frac{\partial^3 P}{\partial V^3} \right|_{V_c, T_c} (V - V_c)^3 + \dots$$

$$\Delta P \propto (\Delta V)^3 = (\Delta V)^\delta$$

$$\delta = 3$$



$$V_g - V_l \propto (T_c - T)^{\frac{1}{2}}$$

$$\beta = \frac{1}{2}$$

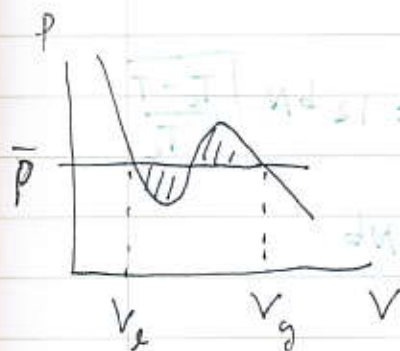
asymptotic result near the critical point.

why.

Date

$$\bar{P} = - \frac{aN^2}{V^2} + \frac{Nk_B T}{V - Nb}$$

Maxwell construction



$$P = - \frac{aN^2}{V^2} + \frac{Nk_B T}{V - Nb}$$

$$\int_{V_l}^{V_g} (P - \bar{P}) dV = 0$$

$$\int_{V_l}^{V_g} \left(- \frac{aN^2}{V^2} + \frac{Nk_B T}{V - Nb} - \bar{P} \right) dV = \left[\frac{aN^2}{V} + Nk_B T \ln(V - Nb) - \bar{P}V \right]_{V_l}^{V_g}$$

$$\Rightarrow \frac{aN^2}{V_g} + Nk_B T \ln(V_g - Nb) + \frac{aN^2}{V_g} - \frac{Nk_B T}{V_g - Nb} \cdot V_g =$$

same exp with V_g replaced by V_l

$$\ln(V - Nb) + \frac{2aN}{k_B T V} - \frac{V}{V - Nb} = f(V)$$

$$f(V_g) = f(V_l)$$

$$V_g = V_c + x$$

$$V_g - V_l = 2x$$

$$V_l = V_c - x$$

$$f(V_g) = f(V_c) + f'(V_c)x + \frac{1}{2}f''(V_c)x^2 + \frac{1}{6}f'''(V_c)x^3 + \dots$$

$$f(V_l) = f(V_c) + f'(V_c)(-x) + \frac{1}{2}f''(V_c)x^2 + \frac{1}{6}f'''(V_c)(-x)^3 + \dots$$

$$f(V_g) = f(V_l) \Rightarrow 2f'x + \frac{1}{3}f'''x^3 = 0$$

$$2f' + \frac{1}{3}f'''x^2 = 0$$

$$\frac{1}{2} = \frac{1}{2} \quad \frac{1}{2} (T_c - T) \approx Y - Y_c$$

near the critical point
using Taylor's series

Date

entropy change near T_c

$$\frac{dP}{dT} = \frac{\Delta S}{\Delta V}$$

$$\Delta V = 2x = 12 b N \sqrt{\frac{T_c - T}{T_c}}$$

$$\frac{dP}{dT} = \frac{N K_B}{V - N b} \approx \frac{N K_B}{2 b N} = \frac{K_B}{2 b}$$

$$V_c = 3 N b$$

$$\begin{aligned} \text{so } \Delta S &= \frac{N K_B}{2 b N} \cdot 12 b N \sqrt{\frac{T_c - T}{T_c}} \\ &= 6 K_B N \sqrt{\frac{T_c - T}{T_c}} \end{aligned}$$

$$P = \frac{N k_B T}{V - N b} = \frac{N k_B T}{V} + (V - N b) \frac{N k_B T}{V^2} + \dots$$

use the binomial expansion

$$(V)^{-1} = \frac{1}{V} = \frac{1}{V} \left(1 - \frac{N b}{V} + \dots \right)$$

$$x = V - N b$$

$$x + N b = V$$

$$(x)^{-1} = (V)^{-1}$$

$$x = V - N b$$

$$x^{-1} = (V - N b)^{-1} = \frac{1}{V} \left(1 + \frac{N b}{V} + \dots \right)$$

$$\frac{1}{x} = \frac{1}{V - N b} = \frac{1}{V} \left(1 + \frac{N b}{V} + \dots \right)$$

$$= \frac{1}{V} + \frac{N b}{V^2} + \dots$$

$$= \frac{1}{V} + \frac{N b}{V^2}$$

$$= \frac{1}{V} + \frac{N b}{V^2}$$

$$f' = \frac{1}{V_c - Nb} - \frac{2aN}{k_B T V_c^2} - \frac{1}{V_c - Nb} + \frac{V_c}{(V_c - Nb)^2}$$

$$= \frac{-2aN}{k_B T (Nb^2 N^2)} + \frac{3bN}{(2bN)^2}$$

$$= -\frac{2}{9} \frac{a}{k_B T b^2 N} + \frac{3}{4} \frac{1}{bN}$$

$$k_B T_c = \frac{8}{27} \frac{a}{b}$$

$$= -\frac{2}{9 k_B T} \left(\frac{27}{8} k_B T_c \right) \frac{1}{bN} + \frac{3}{4} \frac{1}{bN}$$

T close to T_c

$$= -\frac{3}{4} \frac{T_c}{T} \frac{1}{bN} + \frac{3}{4} \frac{1}{bN} = \frac{3}{4bN} \left(1 - \frac{T_c}{T} \right) \approx \frac{3}{4bN} \frac{(T - T_c)}{T_c}$$

$$\lambda(T - T_c) + f''' X^2 = 0$$

$$X \propto \sqrt{T_c - T}$$

as expected

$$f''' = \frac{5}{8b^3 N^3} - \frac{4a}{27b^4 T N^3} \approx \frac{1}{8(bN)^3}$$

$$2 \cdot \frac{3}{4bN} \left(\frac{T - T_c}{T_c} \right) + \frac{1}{3} \frac{1}{8(bN)^3} X^2 = 0$$

$$2 \cdot 3^2 (bN)^2 \left(\frac{T - T_c}{T_c} \right) + X^2 = 0$$

$$X = 6(bN) \cdot \sqrt{\frac{T_c - T}{T_c}}$$

$$= 2V_c \sqrt{\frac{T_c - T}{T_c}}$$

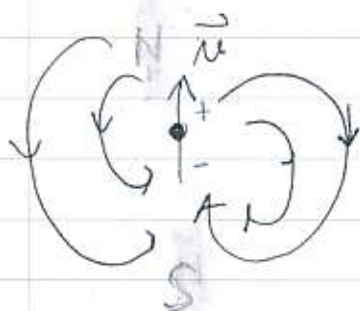
magnetic moment

$$\vec{M} = \frac{1}{2e} \int \vec{r} \times \vec{j}(\vec{r}) dV = \frac{1}{2e} \times (\text{Area})$$

$$\vec{A} = \frac{\mu_0}{4\pi} \frac{\vec{M} \times \vec{r}}{r^3}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{3\hat{r}(\hat{r} \cdot \vec{M}) - \vec{M}}{r^3}$$

$$\text{energy } U = -\vec{\mu} \cdot \vec{B}$$



$\vec{B}$: magnetic induction

$$\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M} \quad \text{magnetic field}$$

↑ magnetic permeability of the vacuum

$$\vec{J} = \vec{L} + \vec{S}$$

27 Feb 07

atoms/ions has magnetic moments (unpaired electron has spin $\vec{S}$
electrons has orbital angle momentum $\vec{L}$)

$$\vec{\mu} = -g \mu_B \vec{J} / \hbar$$

↑ Landé factor

$$g(JLS)$$

$$\mu_B = \frac{e \hbar}{2 m_e}$$

nucleon μ is 10^3 smaller

$$[\hat{J}_x, \hat{J}_y] = i \hbar \hat{J}_z \quad \text{etc.}$$

$$\hat{J} \times \hat{J} = i \hbar \hat{J}$$

Energy of magnetic moment in a field

$$E = -\vec{\mu} \cdot \vec{B} = g \mu_B m B$$

$$\mu_0 H = B$$

electron spin

$$\vec{S} = \frac{\hbar}{2} \vec{\sigma} \leftarrow \text{Pauli matrix}$$

Zeeeman interaction

spin like interaction with magnetic field

$$g = 1 \text{ for } \vec{L}$$

$$g = 2 \text{ for spin}$$

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\vec{\mu} = -g \mu_0 \vec{J} / \hbar \quad \text{classical}$$

$$\vec{J} = \vec{L} + \vec{S}$$

$$\sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\vec{\mu}_1 \uparrow$$

$$\vec{\mu}_2 \uparrow$$

magnet dipole moment interaction

$$U = \frac{1}{r^3} [\vec{\mu}_1 \cdot \vec{\mu}_2 - 3(\vec{\mu}_1 \cdot \hat{r})(\vec{\mu}_2 \cdot \hat{r})]$$

$$U \approx 10^{-4} \text{ eV} \quad \text{too small for ferro magnetic}$$

Heisenberg model

← exchange interaction

arbitrary for Pauli exclusion principle

$$E = -J \sum_{\langle ij \rangle} \hat{\sigma}_i \cdot \hat{\sigma}_j + g \mu_B B \sum_i \hat{\sigma}_i^z$$

Take only z-component

Ising model

$$E = -J \sum_{\langle ij \rangle} \sigma_i \sigma_j - K \sum_i \sigma_i^z$$

$$\sigma_i = \pm 1$$

X-Y model

$$S = Nk_B \left[\ln 2 - \frac{1}{2} (1+m) \ln(1+m) - \frac{1}{2} (1-m) \ln(1-m) \right]$$

canal

① use results of S & M , eliminate h & β

② microcanonical use $S = k_B \ln \Omega$

Paramagnetism

Date

$\mathcal{H} = -h \sum_i \sigma_i$ No interaction between spins (free spins)

$$Z = \sum_{\sigma_i} e^{-\beta \mathcal{H}} = \sum_{\sigma_1} \sum_{\sigma_2} \dots \sum_{\sigma_N} e^{\beta h \sum_{i=1}^N \sigma_i} = \left[\sum_{\sigma} e^{\beta h \sigma} \right]^N$$

$$= (e^{\beta h} + e^{-\beta h})^N$$

$$F = -\frac{1}{\beta} \ln Z = -\frac{1}{\beta} \ln (e^{\beta h} + e^{-\beta h}) = -\frac{k_B T}{\beta} \ln (2 \cosh \frac{h}{k_B T})$$

$$S = -\frac{\partial F}{\partial T} = k_B N \ln (2 \cosh \frac{h}{k_B T}) + k_B N \frac{2 \sinh \frac{h}{k_B T}}{2 \cosh \frac{h}{k_B T}} \cdot \left(-\frac{h}{k_B T^2} \right)$$

$$= k_B N \ln (2 \cosh \frac{h}{k_B T}) - k_B N \left(\frac{h}{k_B T} \right) \tanh \frac{h}{k_B T}$$

$$U = F + TS = -h N \tanh \frac{h}{k_B T} = -\frac{\partial \ln Z}{\partial \beta} = -\frac{\partial}{\partial \beta} \ln [2 \cosh \beta h]^N$$

total magnetization $= -\tanh(\beta h) \times h N$

$$M = -\frac{\partial F}{\partial h} = -\frac{\partial}{\partial h} \left(-\frac{1}{\beta} \ln Z \right) = \frac{1}{\beta} \frac{\partial}{\partial h} \frac{1}{Z} \frac{\partial}{\partial h} \left[\sum_{\sigma} e^{\beta h \sum_{i=1}^N \sigma_i} \right]$$

$$= \frac{1}{\beta} \frac{1}{Z} \sum_{\{\sigma\}} \left(\beta \sum_{i=1}^N \sigma_i \right) e^{\beta h \sum_{i=1}^N \sigma_i} = \left\langle \sum_{i=1}^N \sigma_i \right\rangle = + \frac{N}{\beta} \frac{\partial \ln (e^{\beta h} + e^{-\beta h})}{\partial h}$$

$$= \frac{N}{\beta} \frac{e^{\beta h} - e^{-\beta h}}{e^{\beta h} + e^{-\beta h}} \cdot \beta = N \tanh(\beta h)$$

$F(T, h)$

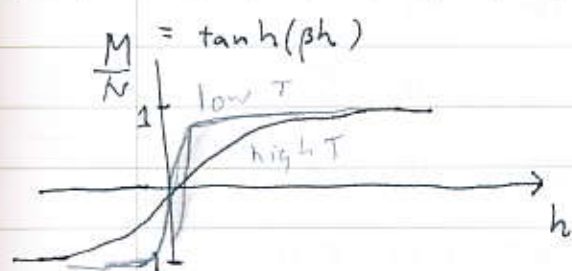
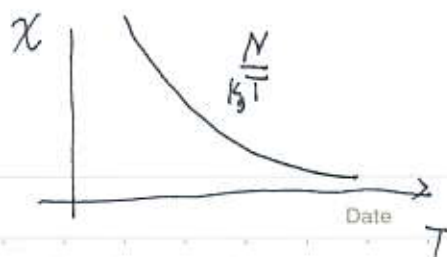
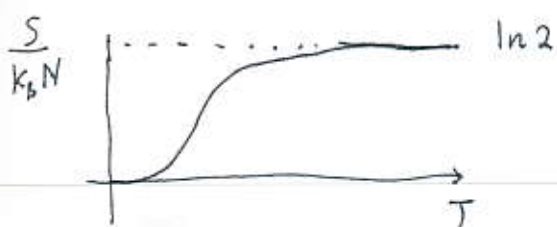
$$dF = \frac{\partial F}{\partial T} dT + \frac{\partial F}{\partial h} dh = -S dT - M dh$$

magnetic susceptibility

compressible fluid (gas)

$$(-S dT - p dV)$$

$$\chi \stackrel{\text{def}}{=} \frac{dM}{dh} \bigg|_{h=0} = N \frac{1}{\cosh^2(\beta h)} \cdot \beta \bigg|_{h=0} = \frac{N}{k_B T} \quad \text{for Curie (Weiss) law}$$



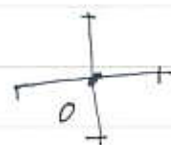
Read chap. 14 & 15
of K. Huang

Mean-field theory of paramagnets $\rightleftharpoons$ ferromagnets

$$H = -J \sum_{\langle ij \rangle} \sigma_i \sigma_j - h \sum_i \sigma_i$$

look at one fixed spin, say $i=0$

$$H_0 = -J \sum_{\langle 0j \rangle} \sigma_0 \sigma_j - h \sigma_0$$



$\langle \sigma_j \rangle \leftarrow$ replace actual value by average

$$= -(J \sum_j \langle \sigma_j \rangle + h) \sigma_0 = -(J q m + h) \sigma_0$$

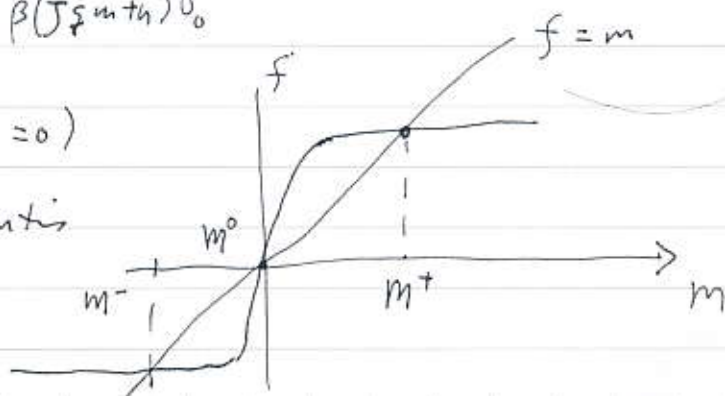
$$\langle \sigma_j \rangle = \langle \sigma_0 \rangle = m = \frac{M}{N}$$

$$m = \langle \sigma_0 \rangle = \frac{\sum_{\sigma_0 = \pm 1} \sigma_0 e^{\beta(J q m + h) \sigma_0}}{\sum_{\sigma_0 = \pm 1} e^{\beta(J q m + h) \sigma_0}} = \tanh(\beta(J q m + h))$$

solution for m (when $h=0$)

$m^0 = 0$ is always a solution

$m^{\pm} \neq 0$ if T is small



comparison of $k_B T_c / J$ on hypercubic lattice (a Hückel)

d	1	2	3	4	5
MF	2	4	6	8	10
Exact	0	2.27	4.5	6.68	8.78

Interpretation by diagrammatic expansion for vector spin models

$$\frac{k_B T_c}{J} = \left(d - \frac{1}{2} - \frac{1}{3d} + \dots \right)^2$$

$$= \left(9 - \frac{1}{2} - \frac{1}{3} \right)^2$$

or $d = 9$ for $d = 3$ and $d = 2$ respectively

$$T_c = \frac{J}{k_B} \left(d - \frac{1}{2} - \frac{1}{3d} + \dots \right)^2$$

or $d = 9$ for $d = 3$ and $d = 2$ respectively

$$T_c(d = 3) = \frac{J}{k_B} \left(9 - \frac{1}{2} - \frac{1}{3} \right)^2 = \frac{J}{k_B} \left(8 \frac{1}{6} \right)^2 = \frac{J}{k_B} \left(\frac{49}{9} \right)$$

$$\frac{M}{N} = m = \langle S_i \rangle = \langle S_i^z \rangle$$

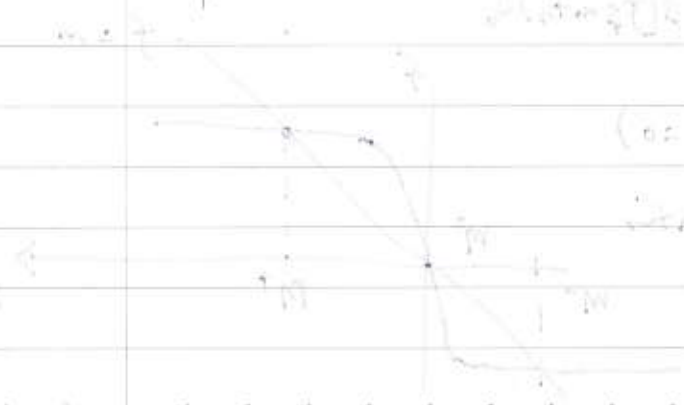
$$T_c(d = 3) = \frac{J}{k_B} \left(\frac{49}{9} \right)$$

$$T_c(d = 2) = \frac{J}{k_B} \left(\frac{49}{9} \right)$$

(or $d = 3$ and $d = 2$ respectively)

Stable or unstable? $d = 3$ and $d = 2$

Value of T_c for $d = 3$ and $d = 2$



critical T $\tanh(\beta J q m) \approx \beta J q m$

we get intersect if slope of tanh is > 1 Date

i.e. $(\beta_c J q) = 1$, or $\frac{J q}{k_B} = T_c$

Value of spontaneous magnetization

$$\tanh x = x - \frac{x^3}{3} + \dots$$

$$m = \tanh(\beta(q J m + h))$$

$$= \beta q J m - \frac{1}{3} (\beta q J)^3 m^3 + \dots \quad m \text{ small}$$

$$1 = \beta q J - \frac{1}{3} (\beta q J)^3 m^2 \quad (\tanh x)' = \frac{1}{\cosh^2 x}$$

$$\beta q J - 1 = \frac{1}{3} (\beta q J)^3 m^2$$

$$m^2 = \frac{3}{(\beta q J)^3} (\beta q J - 1)$$

$$= \frac{3}{(\beta q J)^3} \left(\frac{q J}{k_B T} - 1 \right)$$

$$= \frac{3}{(\beta q J)^3} \left(\frac{T_c - T}{T} \right)$$

$$m \propto (T_c - T)^{\frac{1}{2}} \quad (T \rightarrow T_c) \quad \beta = \frac{1}{2}$$

susceptibility

exact curve is given by $\beta = \frac{1}{k_B T} = \frac{1}{2 q J m} \ln \frac{1+m}{1-m}$

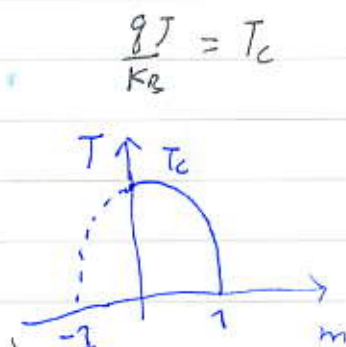
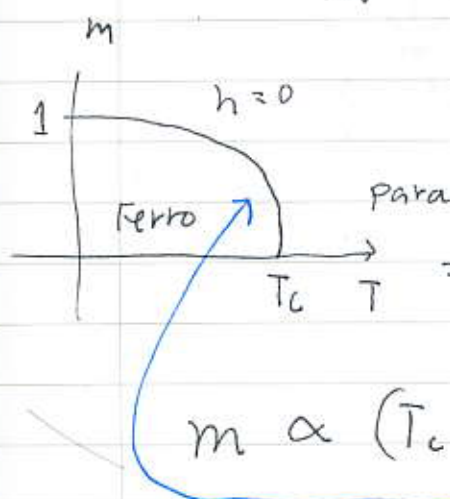
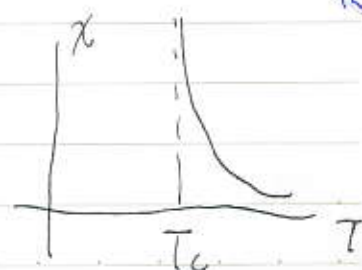
$$\chi = \frac{\partial m}{\partial h} = \frac{1}{\cosh^2(\beta(q J m + h))} \left[\beta q J \frac{\partial m}{\partial h} + \beta \right] \quad \left| \begin{array}{l} T > T_c \quad m=0 \\ h=0 \end{array} \right.$$

$$\chi = \beta q J \chi + \beta$$

$$\chi = \frac{\beta}{1 - \beta q J} = \frac{1}{1 - \frac{T_c}{T}} = \frac{T}{T - T_c}$$

$$= 6 |T - T_c|^{-\gamma} \quad \gamma = 1$$

what is χ for $T < T_c$



$$h = -mgJ + \frac{1}{\beta} \operatorname{arctanh}(m)$$

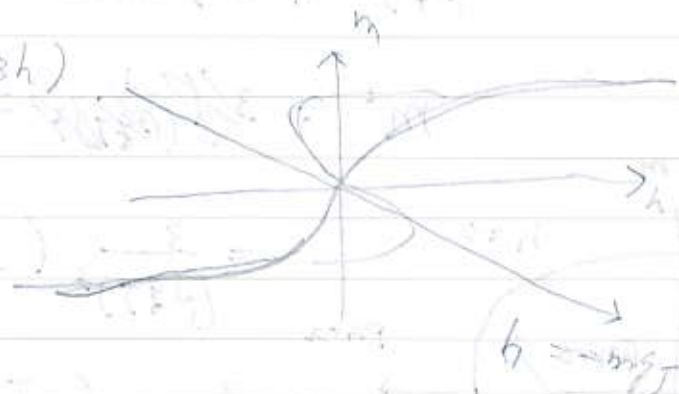
$$h \rightarrow 0 \quad k_B T = \frac{mgJ}{\operatorname{arctanh}(m)}$$

$$y = \operatorname{arctanh}(x) = \tanh^{-1}(x) = \frac{1}{2} \ln \frac{x+1}{x-1} \quad |x| < 1$$

$$\tanh(x) = x$$

$$h = k_B T \tanh(\beta h) \quad (1)$$

$$\rightarrow m = \tanh(\beta h)$$



$$h = -mgJ$$



$$h = 0$$

$$\frac{h}{m} = \frac{-mgJ}{m} = -gJ$$

$$\left[\frac{h}{m} = \frac{-mgJ}{m} \right]$$

$$\frac{h}{m} = \frac{-mgJ}{m} = -gJ$$

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critical isotherm

$$T = T_c$$

Date

$$m = \tanh(\beta_c (J m + h))$$

$$= \tanh(m + \beta_c h)$$

$$= m + \beta_c h - \frac{1}{3} (m + \beta_c h)^3$$

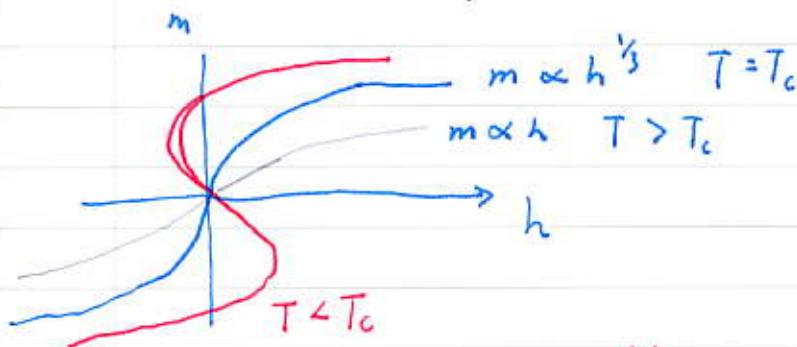
$$\beta_c h = \frac{1}{3} (m + \beta_c h)^3$$

$$\approx \frac{1}{3} m^3$$

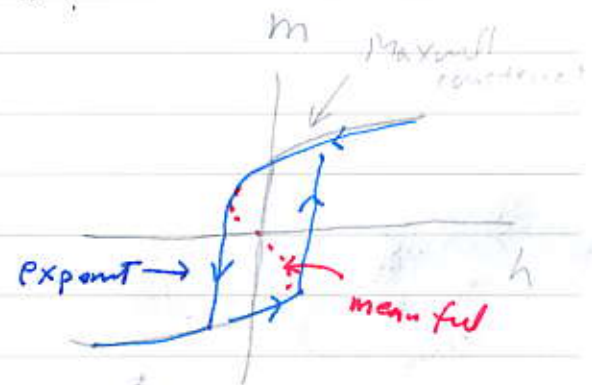
$$\delta = 3$$

m small
 h small
 $T = T_c$

$$h \sim m^3$$



Maxwell construction is needed



hysteresis

$$F = U - TS$$

$$dU = dF + T dS + S dT$$

$$= -S dT - M dh + S dT + T dS = T dS - M dh$$

Landau theory

magnetic system

"Helmholtz" free energy

$$F_{(T,h)} = -\frac{1}{\beta} \ln Z$$

$$Z = \sum_{\{\sigma_i\}} e^{-\beta H}$$

$$dF = -S dT - M dh$$

$$F = F(T, h)$$

"Gibbs" free energy

$$M = N m$$

$$G = F + M h = G(T, M)$$

order parameter

$$dG = -S dT - M dh + M dh + h dM = -S dT + h dM$$

□ G is even in M .

① sign of M is same as h
 $(m \rightarrow -m, h \rightarrow -h)$

Assumption 1

so $M h(M)$ is even in M

$$F = -\frac{1}{\beta} \ln 2 \cosh(\beta(\sqrt{2}m + h)) \leftarrow \text{even function in } m \quad 147$$

at least in MF G^R is even in M , but should be γ true.

□ assumption 2

$$G(T, M) = G(T, 0) + A(T)M^2 + B(T)M^4 + \dots$$

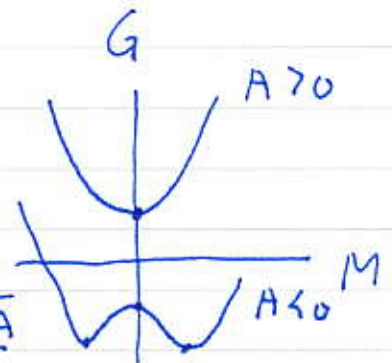
M small
near critical pt

G is analytic in M (order parameter)

$$h = \frac{\partial G}{\partial M} = 2AM + 4BM^3$$

$$\text{At } h=0 \quad (2A + 4BM^2)M = 0$$

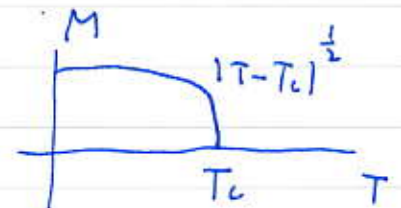
$$M=0 \text{ or } M = \pm \sqrt{\frac{-A}{2B}}$$



□ assumption 3

$$\text{Let } A = a(T - T_c), \quad a > 0$$

$$\rightarrow M \propto (T_c - T)^{\frac{1}{2}} \quad T < T_c$$



$$\beta = \frac{1}{2}$$

at T_c , $A=0$

$$\text{so } h = 4BM^3$$

$$\delta = 3$$

susceptibility

$$\chi_T = \frac{\partial M}{\partial h} \bigg|_T = \frac{\partial h}{\partial M} \bigg|_T^{-1} = \frac{\partial^2 G}{\partial M^2} \bigg|_T^{-1}$$

$$\chi_T^{-1} = 2A + 12BM^2 \bigg|_{T > T_c} = 2A$$

$$\chi_T = \frac{1}{2a(T - T_c)} \propto |T - T_c|^{-\gamma}$$

$$T < T_c \quad h=0$$

$$BM^2 = -\frac{1}{2}A$$

$$\chi_T^{-1} = 2A + \frac{12}{2}A$$

$$= -4A$$

$$= 4a(T_c - T)$$

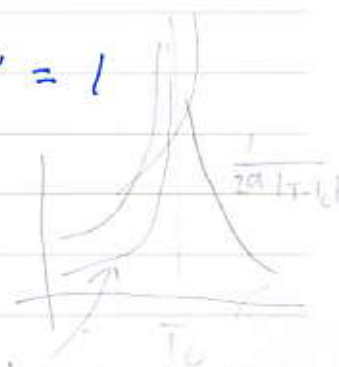
$$M=0$$

$$\text{for } T > T_c$$

$$\gamma = 1$$

When does mean-field is valid?

connection between Van der Waals theory, mean-field,
& Landau theory.



40/27d